AF3060 INVERTER DRIVES

DRIVER - 331X358AAG01, G02

OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62500A

DRIVER ASSEMBLY	ELEM	36D865533AA
	CONN	36C764839AA

PRINTED CIRCUIT CARD DIAGRAMS

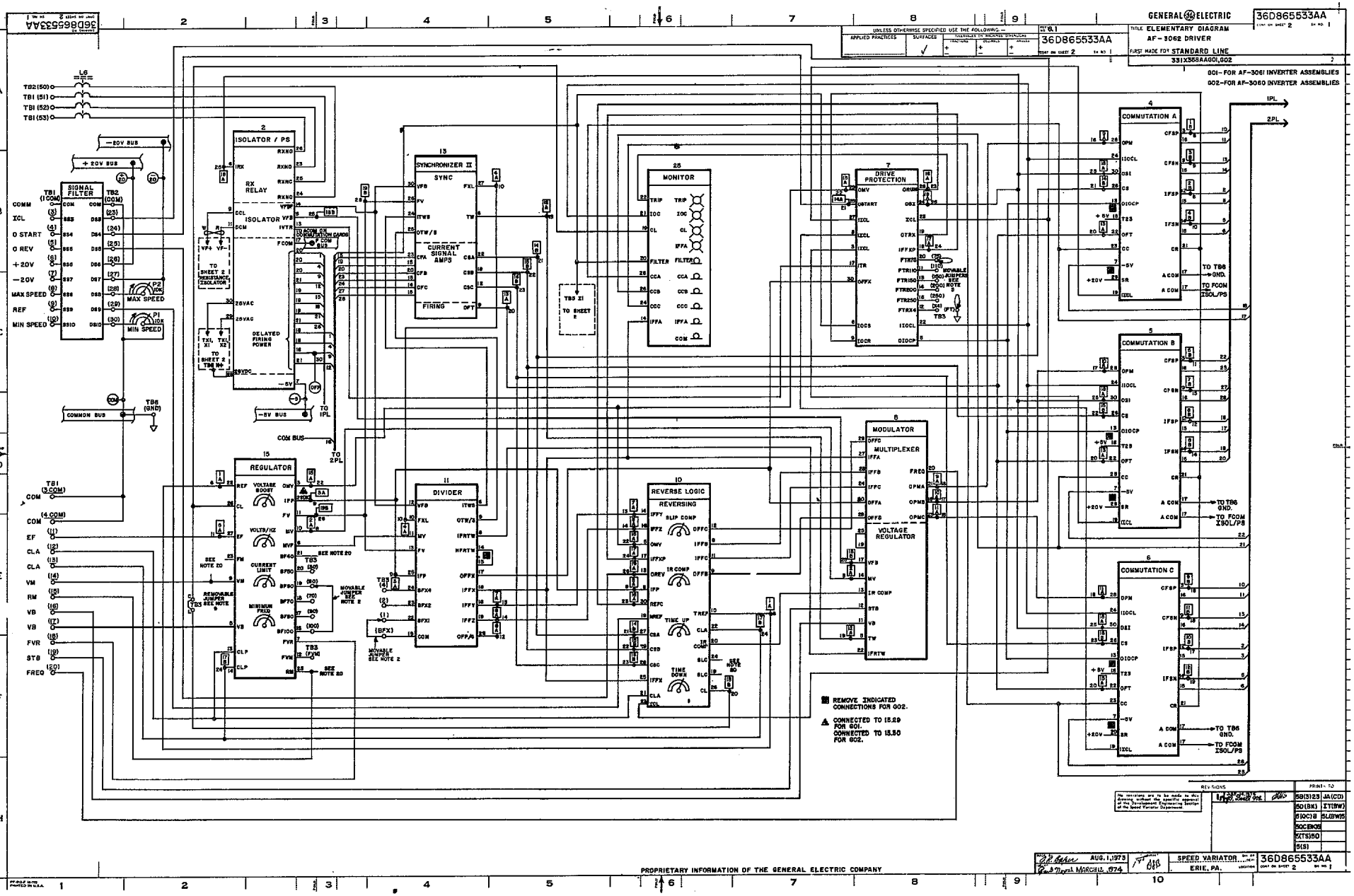
DRIVE PROTECTION	36C762849AC
COMMUTATION	36C764126AE
REGULATOR	36C764136AB
MODULATOR	36C764107AA
SYNCHRONIZER II	36C764108AA
REVERSE LOGIC	36C762855AB
SIGNAL FILTER	36C764868AA
MONITOR	36C762854AA
ISOLATOR/POWER SUPPLY	36C762869BR
5V POWER SUPPLY	36C762870AB
20V POWER SUPPLY	36C762896AB
TEST CARD	36C762858AA
DIVIDER	36C762895AB
POWER INTERFACE	36C764129AA
RESISTANCE ISOLATOR	36C764140AA

ISSUED 9/9/76

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

...k of the General Electric Co.


GENERAL ELECTRIC
36D865533AA
TITLE ELEMENTARY DIAGRAM
AF-3062 DRIVER
FIRST MADE FOR STANDARD LINE
331X368AAG01,G02
601-FOR AF-3061 INVERTER ASSEMBLIES
602-FOR AF-3060 INVERTER ASSEMBLIES
SIGNAL FILTER
ISOLATOR / PS
SYNCHRONIZER II
SYNC
CURRENT SIGNAL AMPS
FIRING
MONITOR
DRIVE PROTECTION
COMMUTATION A
COMMUTATION B
COMMUTATION C
MODULATOR
MULTIPLEXER
VOLTAGE REGULATOR
REGULATOR
DIVIDER
REVERSE LOGIC
REVERSING
TIME UP
TIME DOWN
RX RELAY
DELAYED FIRING POWER
MAX SPEED
MIN SPEED
COMMON BUS
COM BUS
TO 2PL
TO 1PL
REMOVE INDICATED CONNECTIONS FOR G02.
CONNECTED TO 15.29 FOR G01. CONNECTED TO 15.30 FOR G02.
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE, PA.
36D865533AA

601-FOR AF-3061 INVERTER ASSEMBLIES
602-FOR AF-3060 INVERTER ASSEMBLIES

DRIVER NOMENCLATURE

NOMENCLATURE

THE ABBREVIATED NOMENCLATURE USED ON THE DRIVER ELEMENTARY BELOW, A "1" (5 VOLTS) OR A "0" (ZERO VOLTS - PHASE AND THE ABBREVIATION INDICATES A LOGIC "1" SIGNAL ARE NOTED. TO THE MAIN MOTOR LOGIC STATE OR THE LOGIC STATE WHEN THE CORRESPONDING PHASE IN THE POWER INVERTER IS POSITIVE. PHASES A, B, AND C REFER TO THE POWER PHASES. PHASES X, Y, AND Z REFER TO THE DRIVER PHASES ONLY THEY ARE CONNECTED TO A POWER PHASE. THE FOLLOWING RELATION EXISTS BETWEEN THE DRIVER AND POWER PHASES.

DRIVER PHASE	POWER PHASE	
	FORWARD	REVERSE
X	A	A
Y	B	C
Z	C	B

EXAMPLES:

- 1IOCL - INSTANTANEOUS OVERCURRENT LOCKOUT, LOCKS OUT FIRING SIGNAL WHEN THE "1" STATE.
- OFFX - THE FUNDAMENTAL DRIVER FREQUENCY OF PHASE X (POWER PHASE A). THIS SIGNAL IS IN THE "0" STATE WHEN PHASE A IS POSITIVE.

BFX1	BASE FREQUENCY TIMES 1
BFX2	BASE FREQUENCY TIMES 2
BFX4	BASE FREQUENCY TIMES 4
BF40 (80,60,ETC.)	BASE FREQUENCY - 40 HZ (50 HZ, 60 HZ, ETC.)
CSA (B,C)	CURRENT FEEDBACK - PHASE A (B,C)
CFA (B,C)	CURRENT FEEDBACK - PHASE A (B,C)
CFSN	COMMUTATION FIRING SIGNAL - NEGATIVE BUS
CFSP	COMMUTATION FIRING SIGNAL - POSITIVE BUS
CL	CURRENT LIMIT
CLA	CURRENT LIMIT ADJUST
CLP	CURRENT LIMIT POTENTIOMETER
CR	COMMUTATION RECALIBRATION
CSA (B,C)	CURRENT SIGNAL - PHASE A (B,C)
DCL	D C LINK
DCM	D C MIDPOINT
DCP	DCL VOLTAGE POWER
DF	EXTERNAL FREQUENCY
OFFA	FUNDAMENTAL FREQUENCY - PHASE A
1FFA	FUNDAMENTAL FREQUENCY - PHASE A
OFFB	FUNDAMENTAL FREQUENCY - PHASE B
1FFB	FUNDAMENTAL FREQUENCY - PHASE B
1FFC	FUNDAMENTAL FREQUENCY - PHASE C
1FFC	FUNDAMENTAL FREQUENCY - PHASE C
FILTER	FILTER CURRENT FEEDBACK
OFFX	FUNDAMENTAL FREQUENCY - PHASE X
1FFX	FUNDAMENTAL FREQUENCY - PHASE X
1FFXP	A "1" PULSE RELATED TO PHASE X
1FFY	FUNDAMENTAL FREQUENCY - PHASE Y
1FFZ	FUNDAMENTAL FREQUENCY - PHASE Z
1FP	FREQUENCY PULSE
OFF/2	FREQUENCY PULSE DIVIDED BY 2
FM	FREQUENCY MODIFICATION
FREQ	FREQUENCY PULSE READOUT
1FRTW	FREE RUNNING TRIANGLE WAVE
FT	FIRING TRAIN
FTR75(11,,15,,ETC	FREQUENCY TRIP - 75 HZ (11 HZ,15 HZ,ETC.)
FTRX4	FREQUENCY TRIP - TIMES FOUR
FV	FREQUENCY VOLTAGE
FVR	FREQUENCY VOLTAGE READOUT
FXL	FREQUENCY TIMES DC LINK
FVM	FREQUENCY VOLTAGE MODIFICATION
ICL	INSTANTANEOUS CURRENT LIMIT
1FSN	INVERTER FIRING SIGNAL - NEGATIVE BUS
1FSP	INVERTER FIRING SIGNAL - POSITIVE BUS
HFTRW	HIGH FREQUENCY TRIANGLE WAVE
1IOCL	INSTANTANEOUS OVERCURRENT LOCKOUT
RM	REFERENCE MODIFICATION
01OCP	INSTANTANEOUS OVERCURRENT PULSE
IOCR	INSTANTANEOUS OVERCURRENT RECALIBRATION
IOCS	INSTANTANEOUS OVERCURRENT SIGNAL
I R COMP	MOTOR RESISTANCE COMPENSATION
MMV	MINIMUM VOLTAGE
MV	MODULATOR VOLTAGE
MVCP	REFERENCE
OPWM (B,C)	PULSE MODULATION SIGNAL - PHASE A (B,C)
REF	REFERENCE
REFC	REFERENCE CLAMP
REVR	REVERSE
RSLM	RUN SIGNAL RECEIVED
RSX	RX RELAY - PICKS UP ON "1" SIGNAL
RXNC	RX RELAY - NORMALLY CLOSED CONTACTS
RXNO	RX RELAY - NORMALLY OPEN CONTACTS
SSI	STEP INDICATOR
TTWS	TRIANGLE WAVE SYNCH
TW/3	TRIANGLE WAVE DIVIDED BY 3
T2S	2S MICRO SECOND TIMER
SLC	SLIP COMPENSATION
USTART	START
TTR	DRIVE TRIGGER "1" PULSE
TVCP	TIMED REFERENCE
OTRX	TRIP EXECUTE
TW	TRIANGLE WAVE
VB	VOLTAGE BOOST
VFB	VOLTAGE FEEDBACK
VTR	VOLTAGE TRIP
VR	STEEP RAMP
VSB	STABILITY
VM	VOLTAGE MODIFICATION

DRIVER NOTES

1. THE NUMBERS ADJACENT TO THE CARD BLOCKS REFER TO THE LOCATION OF THE CARD IN THE DRIVER RACK.
2. BASE FREQUENCY IS THE FREQUENCY AT WHICH THE DRIVE REACHES FULL VOLTAGE. THE BASE FREQUENCY IS SELECTED BY A BASE FREQUENCY JUMPER CONNECTED TO TB3(100) AND A BASE FREQUENCY MULTIPLIER JUMPER CONNECTED TO TB3(FX). THE JUMPERS SHOWN ARE FOR A 60 HZ DRIVE. SEE THE TABLE BELOW FOR PROPER PLACEMENT OF THESE JUMPERS. CHECK DRIVE ELEMENTARY DIAGRAM FOR PROPER PLACEMENT OF THIS JUMPER. CAUTION: IMPROPER PLACEMENT OF THIS JUMPER MAY RESULT IN DRIVE MALFUNCTION OR DAMAGE. DO NOT MAKE ANY CHANGES WITHOUT CONSULTING GENERAL ELECTRIC COMPANY.

TB3 JUMPER	HERTZ					
MULTIPLIER	BFX 100	(40)	(50)	(60)	(70)	(80)	(100)
	(1)	40	50	60	70	80	100
	(2)	80	100	120	140	160	200
	(4)	160	200	240	280	320	

002 DRIVERS WILL HAVE 1/2 THE INDICATED FREQUENCY

3. JUMPERS ANCHORED TO TB3(75) ARE (44) SELECT THE FREQUENCY TRIP AT ARE CONNECTED AS SHOWN IN THE TABLE BELOW. CHECK THE DRIVE ELEMENTARY DIAGRAM FOR PROPER PLACEMENT OF THESE JUMPERS. WARNING: IMPROPER PLACEMENT OF THESE JUMPERS MAY RESULT IN DRIVE MALFUNCTION OR PRESENT AN EQUIPMENT OR PERSONNEL HAZARD. DUE TO MOTOR VARIATIONS, DO NOT MAKE ANY CHANGES WITHOUT CONSULTING GENERAL ELECTRIC COMPANY.

TB3(75) JUMPER PLACEMENT	75	110	150	230	250
FREQUENCY TRIP (HZ)	75	110	15	230	8
WITH TB3(FT) JUMPERED TO (44)	30	440	60	80	100

4. SEVERAL DRIVER CIRCUITS ARE WIRED TO THE MONITOR CARD TEST CONNECTORS TO ENABLE EXTERNAL CONNECTIONS OF TEST PROBES.

   WHEN LIT, THE LIGHTS ON THE MONITOR CARD MAKE THE FOLLOWING INDICATIONS.

LIGHT	INDICATION
TRIP	DRIVE TRIP DUE TO UNDERVOLTAGE, OVERVOLTAGE OR HIGH FREQUENCY (IOC WILL ALSO LIGHT)
IOC	SHOOT THROUGH OR OTHER MAJOR POWER SYSTEM FAULT
CL	DRIVE IS IN CURRENT LIMIT
1FFA	PHASE A FREQUENCY SIGNAL

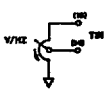
5. TEST RECEPTACLES "A" AND "B" SHARE ONE TEST CARD WHICH HAS A HOME LOCATION AT "A", POSITION NO.25. THESE RECEPTACLES AND TEST CARD ARE USED TO MONITOR THE DRIVE CIRCUITS WIRED TO TB24.

   A WIRE PERMANENTLY CONNECTED TO TAB 28 OF BOTH TEST RECEPTACLES HAS A MOVEABLE TEST CONNECTOR ON THE OTHER END WHICH MAY BE CONNECTED TO ANY POINT IN THE BACK PLANE WIRING. THIS WIRE IS CONNECTED TO THE "SEL" TEST CONNECTOR FOR MONITORING ANY POINT DESIRED.

   SINCE MANY OF THE TEST POINTS ARE LOGIC SIGNALS, PROPERLY SHIELDED TEST LEADS SHOULD BE USED AT ALL TIMES.

6. THE "MIN. FREQ." POTENTIOMETER P258 ON THE REGULATOR CARD ADJUSTS THE MINIMUM FREQUENCY. A REFERENCE SETTING BELOW THE MINIMUM FREQUENCY WILL REDUCE THE OUTPUT VOLTAGE BUT NOT THE FREQUENCY. THIS WILL PROVIDE SMOOTH STARTUP AND REVERSING.
7. THE "VOLTS/HERTZ VERNIER" POTENTIOMETER P267 ON THE REGULATOR CARD ADJUSTS THE VOLTS PER HERTZ OF THE INVERTER WITHIN ±10% OF NOMINAL.

   AN EXTERNAL VOLTS PER HERTZ POTENTIOMETER MAY BE ADDED AS INDICATED BELOW. THIS POTENTIOMETER WILL ALLOW AN EXTERNAL VOLTS/HZ ADJUSTMENT OF 0:1 AS AN INCREASE OVER THE INTERNAL SETTING.



4. IF A VOLTAGE BOOST FUNCTION IS PROVIDED THE "VOLTAGE BOOST" POTENTIOMETER P265 ON THE REGULATOR CARD CAN BE USED TO ADD A FIXED VOLTAGE TO THE INVERTER IN OPEN LOOP FREQUENCY. AN EXTERNAL VOLTAGE BOOST POTENTIOMETER MAY BE PROVIDED.
9. THE "CURRENT LIMIT" POTENTIOMETER P264 ON THE REGULATOR CARD ADJUSTS THE CURRENT LIMIT BETWEEN 5% AND 150% OF RATED DRIVE CURRENT. AN ADDITIONAL EXTERNAL CURRENT LIMIT RHEOSTAT MAY BE PROVIDED TO DECREASE THE CURRENT LIMIT BELOW 150% CURRENT LIMIT MAY BE REDUCED IMPERATIVE BY REMOVING THE "CL" JUMPER ON TB6.
10. THE "I.R. COMP." POTENTIOMETER P475 ON THE REVERSE LOGIC CARD MAY BE ADJUSTED TO COMPENSATE FOR VOLTAGE LOSS DUE TO MOTOR RESISTANCE. IF NOT USED, TURN FULL CCW.
11. IF SLIP COMPENSATION IS PROVIDED THE "SLIP COMP." POTENTIOMETER P475 ON THE REVERSE LOGIC CARD MAY BE ADJUSTED TO COMPENSATE FOR SLIP IN MOTOR SPEED DUE TO LOADING.
12. FOR A VOLTAGE REFERENCE THE "TIME UP" P477 AND "TIME DOWN" P477 POTENTIOMETER ON THE REVERSE LOGIC CARD ADJUST THE ACCELERATION AND DECELERATION RATES RESPECTIVELY. THE TIME FOR THE DRIVE TO RESPOND TO A 10V STEP CHANGE IN REFERENCE MAY BE VARIED BETWEEN 2.0 SEC AND 32 SEC.
13. THE FAST CURRENT LIMIT SETTING IS FIXED AND SET AT A PREDETERMINED VALUE APPROXIMATELY 2.5% OF PEAK RATED CURRENT. THIS IS A "DEEP RUNNING" LIMIT THAT REDUCES VOLTAGE BUT DOES NOT TRIP THE DRIVE.
14. CONNECTOR PIN NUMBERS FOR 1PL AND 2PL ARE SHOWN AS FOLLOWS



THE HEAVY LINES REPRESENT THE CABLES LEADING TO THE FEMALE CONNECTORS 1PL AND 2PL MOUNTED NEAR THE BOTTOM OF THE CARD RACK.

15. (41) — INDICATES TERMINAL BOARD NUMBER



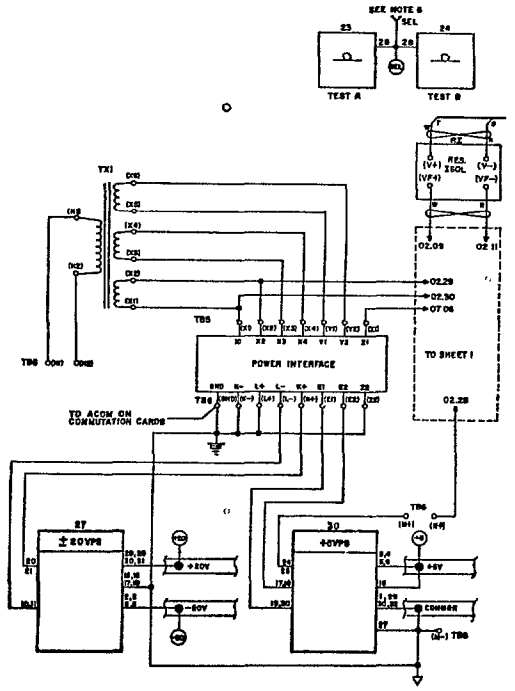
INDICATES TAB MARKER ON CARD WHICH CORRESPONDS TO MATCHING RECEPTACLE NUMBER

16. POTENTIOMETER ARROWS INDICATE THE WIPER DIRECTION ON THE ELEMENTARY DIAGRAM AS THE POTENTIOMETER SHAFT IS ROTATED CLOCKWISE.
17.  INDICATES TEST CONNECTOR ON TEST CARD.
18.  NUMBER IN SYMBOL CORRESPONDS TO INDICATOR POSITION ON THE TEST CARD. LETTER IN SYMBOL CORRESPONDS TO POSITION TEST CARD IS LOCATED. NUMBER BELOW THE SYMBOL INDICATES THE TAB NUMBER IN THE TEST CARD
19.  SYMBOL FOR TEST CONNECTOR.
20. THE FOLLOWING MODIFICATIONS MAY BE MADE USING RED NYLON JACKETED #24 WIRE.

    15 09 - 15 25 DOUBLES REFERENCE TO VOLTAGE GAIN SUCH AS FOR WIDE CONSTANT HORSE POWER RANGE

    15.23 - 15 25 DOUBLES REFERENCE TO FREQUENCY GAIN SUCH AS FOR HIGH FREQUENCY DRIVES.

    15.18 - 15 23 FOR 40 HZ BASE FREQUENCY (WITH BASE FREQUENCY JUMPER SELECTION ON 100).



WIRE TABLE		
BUS BAR CONNECTIONS		
TAB	CARDS CONNECTED	CIRCUIT
02	02, 04, 05, 06, 07, 08, 10, 11, 13, 15	-20V
04	02, 04, 05, 06, 07, 08, 10, 11, 13, 15	+5V
07	02, 04, 05, 06, 07,	-5V
31	02, 04, 05, 06, 07, 08, 10, 11, 13, 15	+20V
32	02, 04, 05, 06, 07, 08, 10, 11, 13, 15	COM
17	02, 04, 05, 06	F COM

BACK PLANE WIRING
#24 WHITE

02.05-08.17-24.25
02.06-04.30-05.30-06.30-07.24-23.25
02.13-07.17
02.14-11.12-13.30
02.18-23.30-24.30

04.03-24.06
04.09-24.09
04.12-24.08
04.13-05.13-06.13-07.05
04.14-24.10
04.18-04.04 M
04.19-07.27
04.21-05.21-06.21-07.09
04.22-05.22-06.22-13.09-23.20
04.23-25.28
04.24-05.24-06.24-07.22-25.21
04.26-10.27-13.22-24.21
04.28-08.21-23.16
04.29-04.31 R

05.03-24.11
05.09-24.13
05.12-24.12
05.14-24.14
05.18-05.04 R
05.19-07.03
05.23-25.26
05.26-10.29-13.18-24.22
05.28-08.18-23.17
05.29-05.31 M

06.03-24.16
06.09-24.18
06.12-24.17
06.14-24.19
06.18-06.04 R
06.19-07.08
06.23-25.24
06.26-10.28-13.12-24.23
06.28-08.23-23.18
06.29-06.31 M

07.06-25.30
07.18-10.17-23.24
07.19-25.22
07.23-10.23
07.25-23.21
07.26-10.30-23.23
07.29-10.05-15.03-23.22
07.30-08.30-11.17

08.05-13.08-23.19
08.13-10.20
08.14-15.10-23.08
08.19-08.25
08.22-11.08
08.24-10.11
08.26-10.08
08.27-10.25-11.16-25.14
08.28-10.09
08.29-10.12

10.06-11.25-23.09-15.29 ▲
10.06-11.25-23.09-15.30 ▲
10.10-15.22-23.06
10.13-23.26
10.14-11.18-23.13
10.16-11.19-23.14
10.21-15.13
10.22-15.14-24.24
10.26-24.20

11.06-13.24
11.09-13.25
11.10-13.27-23.10
11.11-15.06
11.13-13.26-15.11-24.26
11.14-11.15 R
11.29-23.12

15.26-25.19
15.27-23.11

23.28-24.28

#24 WHITE
+5V

30.03-10.04
30.04-11.04
30.05-13.04
30.05-25.04-24.04-23.04
30.06-15.04
30.06-30.16

-5V

02.07-23.07-24.07-25.07

±20V

27.28-07.31
27.29-08.31
27.30-10.31
27.31-11.31
27.31-24.31-23.31

-20V

27.02-07.02
27.03-08.02
27.05-10.02
27.06-11.02
27.02-24.02-23.02

COMMON

02.17-23.01-24.01-25.01
27.15-05.32
27.16-06.32
27.17-07.32
27.18-08.32

30.01-10.32
30.01-30.27
30.29-11.32
30.30-13.32
30.32-15.32

BACK PANEL WIRING
#14 BLACK

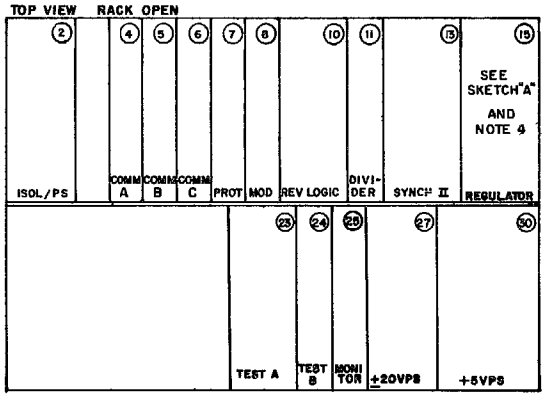
TB1.3COM-TB2.COM-TB6.GND
TB6.GND-TB6.K(MINUS)
TB6.L(PLUS)-TB6.Z2

TX1.H1-TB6.H1
TX1.H2-TB6.H2
TX1.X1-TB5.X1
TX1.X2-TB5.X2
TX1.X3-TB5.X3
TX1.X4-TB5.X4
TX1.X5-TB5.Y1
TX1.X6-TB5.Y2

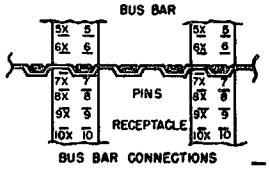
#24 WHITE

P1.02-P1.03-TB6.GND
P1.01-TB2.30
P2.01-P2.02-TB2.28
P2.03-TB2.27

TOP VIEW RACK OPEN



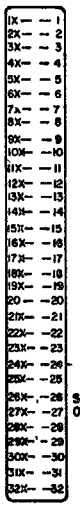
SKETCH "B"
BUS BAR



BUS BAR CONNECTIONS

FACTORY NOTES:
1 ASSEMBLE BUS BAR BEFORE WIRING ALWAYS ASSEMBLE WITH THE BODY OF THE BUS TOWARD THE LOWER NUMBERED SIDE OF THE RECEPTACLE
2 CONNECT WIRES TO TB1, TB2 & TB3 AT LOWER TERMINALS; ALL WIRES TO TB5 & TB6 AT UPPER TERMINALS.
3 *-CRIMP ALL #24 WIRES ON ONE SPADE TERMINAL.
**-CRIMP THREE #24 WIRES ON EACH OF TWO SPADE TERMINALS.
4. TB3 HAS BOTH SCREW AND TERMPOINT CONNECTORS.

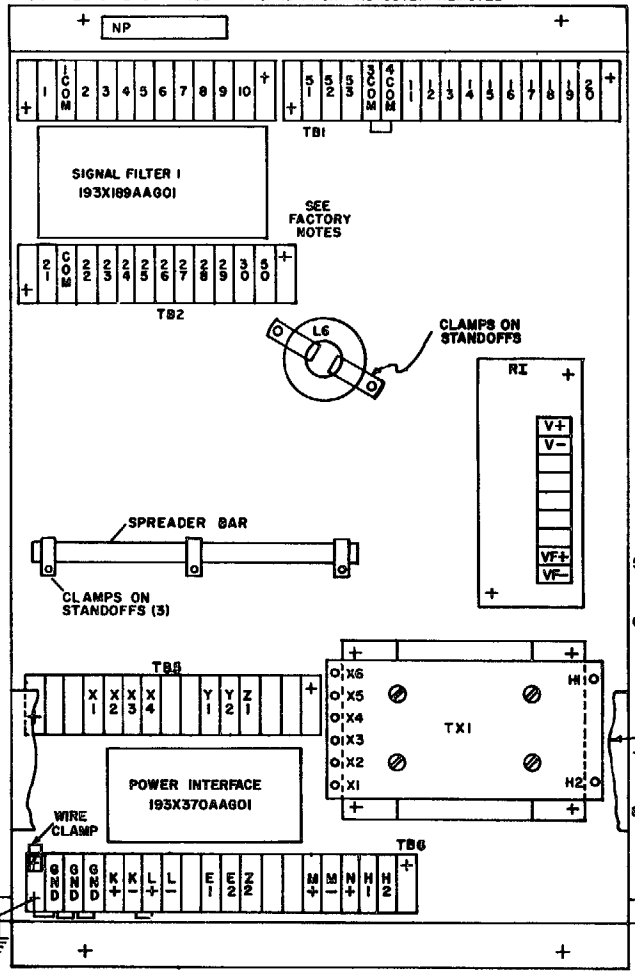
TB3 + 50 60 70 80 100PX 1 2 4 + | + 75 110 150 200 250 FT X4 FVM C L +



SEE NOTE 3 ON SH. 2

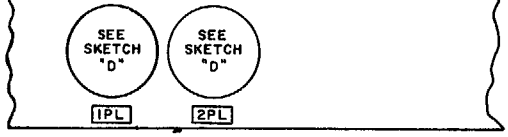
SKETCH "A"
FULL SIZE VIEW OF CARD RECEPTACLE

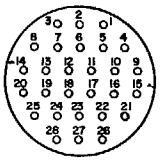
FRONT VIEW OF BACK PANEL WITH CARD RACK AND COVER REMOVED



5. DO NOT BUNDLE WIRES. DISTRIBUTE WIRES ALONG LENGTH OF SPREADER BAR.
6 CUT ALL WIRES APPROXIMATELY ONE INCH LONGER THAN THE SHORTEST POSSIBLE LENGTH.
7. ■ REMOVE INDICATED CONNECTIONS FOR G02
8. ▲ CONNECTED TO 15.29 FOR G01. CONNECTED TO 15.30 FOR G02.

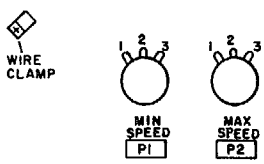
REAR VIEW OF CUTOUT SKETCH "C"





SKETCH "D"

VIEW OF PINNED PANEL OPEN POSITION



G01—FOR AF-3061 INVERTER ASSEMBLIES
G02—FOR AF-3060 INVERTER ASSEMBLIES

TITLE CONNECTION DIAGRAM
AF3062 DRIVER
FIRST MADE FOR STANDARD LINE
331X358AAG01, G02

APPLIED PRACTICES 718A505
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
SURFACES
TOLERANCE OF MACHINED DIMENSIONS DECIMALS FRACTIONS ANGLES

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

ISSUED Feb 6, 1974
Revised MARCH 12, 1974

SPEED VARIATOR
ERIE, PA
DIV OF DEPT
LOCATION

REVISIONS

No revisions may be made to this drawing without approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO
JA(CD) 5(S)
IT(BW) 88(312)3
5L(PW)5 5D(BK)
5QC(B)
5QCB05
5TT250

36C764839AA SH NO 1

36C764839AA REV 1

36C764839AA SH NO 1

36C764839AA SH NO 2

INTERFACE WIRING

#24 WHITE

TB2.50-16 CORE(1 TURN)-J2.24
TB1.51-16 CORE(1 TURN)-J2.25
TB1.52-L6 CORE(1 TURN)-J2.23
TB1.53-L6 CORE(1 TURN)-J2.26

TB1.11-15.27
TB1.12-15.13
TB1.13-10.22
TB1.14-15.09
TB1.15-15.25
TB1.16-08.11
TB1.17-15.05
TB1.18-15.07
TB1.19-08.12
TB1.20-08.20

TB2.23-07.23
TB2.24-07.25
TB2.25-10.13
TB2.26-27.29
TB2.27-27.03
TB2.29-10.19

TB3.50-15.20
TB3.60-15.19
TB3.70-15.18
TB3.80-15.17
TB3.100-15.16
TB3.BFX-11.15

TB3.01-11.22
TB3.02-11.23
TB3.04-11.24
TB3.75-07.20
TB3.110-07.11
TB3.150-07.13
TB3.200-07.14
TB3.250-07.16
TB3.FT-07.15
TB3.X4-07.12
TB3.FVM-15.12
TB3.C-15.26
TB3.L-10.26

TB5.Z1-07.06

TB6.GND-04.17 *
TB6.GND-05.17 *
TB6.GND-06.17 *
TB6.K(PLUS)-27.20 *
TB6.K(PLUS)-27.20 *
TB6.K(PLUS)-27.21 *
TB6.L(MINUS)-27.10 *
TB6.L(MINUS)-27.10 *
TB6.L(MINUS)-27.11 *
TB6.E1-30.19 *
TB6.E1-30.19 *
TB6.E1-30.20 *
TB6.E2-30.17 *
TB6.E2-30.17 *
TB6.E2-30.18 *
TB6 M(PLUS)-30.24 *
TB6.M(PLUS)-30.24 *
TB6.M(PLUS)-30.25 *
TB6.M(MINUS)-30.29 *
TB6.M(MINUS)-30.30 *
TB6.M(MINUS)-30.32 *
TB6.N(PLUS)-02.28 *
TB6.N(PLUS)-02.28 *

TX1.X1-02.30 *
TX1.X1-02.30 *
TX1.X1-02.30 *
TX1.X2-02.29 *
TX1.X2-02.29 *
TX1.X2-02 29 *

#24 RED & WHITE TWISTED

R1.VF(PLUS)-02.09 W
R1.VF(MINUS)-02.11 R

R1.V(PLUS)-2PL.07 W
R1.V(MINUS)-2PL.08 R

PLUG WIRING

#24 WHITE

1PL.01-02.20
1PL.02-04.12
1PL.03-04.15
1PL.04-02.20
1PL.05-04.14
1PL.06-04.15
1PL.09-02.20
1PL.10-04.03
1PL.11-04.16
1PL.12-02.21
1PL.13-04.09
1PL.14-04.16
1PL.15-02.19
1PL.16-05.12
1PL.17-05.15
1PL.18-02.19
1PL.19-05.14
1PL.20-05.15
1PL.21-02.19
1PL.22-05.03
1PL.23-05.16
1PL.26.02.21
1PL.27-05.09
1PL.28-05.16

2PL.01-02.18
2PL.02-06.12
2PL.03-06.15
2PL.04-02.18
2PL.05-06.14
2PL.06-06.15
2PL.09-24.30
2PL.10-06.03
2PL.11-06.16
2PL.12-02.21
2PL.13-06.09
2PL.14-06.16
2PL.15-07.06
2PL.16-02.32
2PL.17-04.23
2PL.18-04.07
2PL.19-13.23
2PL.20-13.15
2PL.21-05.23
2PL.22-15.07
2PL.23-13.20
2PL.24-13.15
2PL.25-06.23
2PL.26-06.07
2PL.27-13.14
2PL.28-13.15

MOVEABLE JUMPER CONNECTIONS

THE FIRST WIRE CONNECTION NUMBER IS THE PERMANENT ONE. THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE.

#24 WHITE

24.28-24.28

THE FIRST WIRE CONNECTION IS THE PERMANENT ONE USING A TERMIPOINT CONNECTION
THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE USING A SPADE TERMINAL

#24 RED

TB3.100-TB3.60, TB3.BFX-TB3.01, TB3.75-TB3.75, TB3 FT-TB3 FT } WIRES SHOULD BE 3.5 INCHES IN LENGTH

■ REMOVE INDICATED CONNECTIONS FOR G02

▲ CONNECTED TO 15.29 FOR G01
CONNECTED TO 15.30 FOR G02

FRONT VIEW OF CARD RACK CLOSED POSITION

		23	24	25	27	30
		193X188AA G01 (SEE NOTE 5)	193X188AA G01	193X184AA G01	193X257AA G01	193X191AB G01
			TEST B	MONI TOR	+20VPS	+5VPS

← LOCATION NUMBERS

2		4	5	6	7	8	10	11	13	15
193X190AF G02		193X369AA G01	193X369AA G01	193X369AA G01	193X178AC G01	193X197AA G01	193X185AA G01	193X195AA G01	193X198AA G01	193X371AA G01
ISOL/PS		COMM A	COMM B	COMM C	PRO TECT ION	MODU LA TOR	REV LOGIC	DIV IDER	SYN CHRONIZER II	REGULATOR

← LOCATION NUMBERS

NOTES:

1. LOCATION NUMBERS AT THE TOP OF EACH CARD INDICATE THE SPACE NUMBER THAT THE CARD'S RECEPTACLE OCCUPIES. THE CARD OCCUPIES THAT SAME SPACE AND AN INTEGRAL NUMBER OF SPACES TO THE LEFT. THE SPACE UNIT REQUIREMENT FOR A CARD CAN BE DETERMINED BY SUBTRACTING THE SPACE NUMBER OF THE CARD TO THE LEFT.

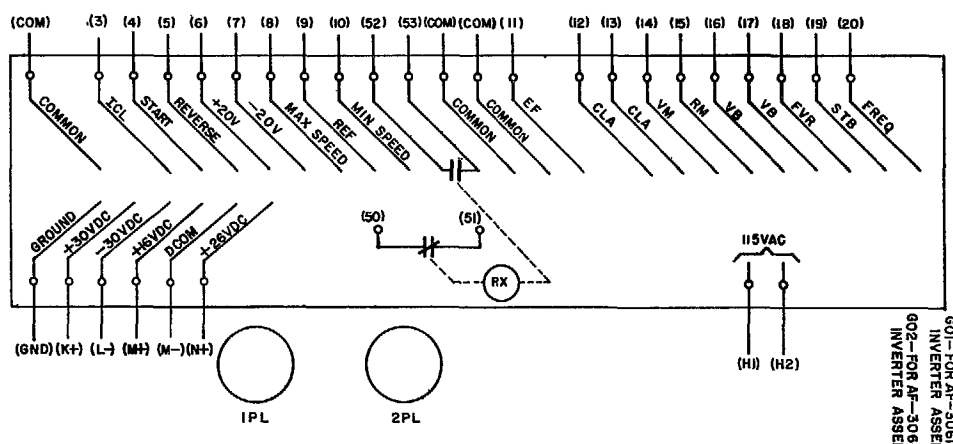
2. THE WIRE TABLE LISTS ALL WIRE CONNECTIONS IN THE DRIVER. A CONNECTION IS INDICATED BY A CARD LOCATION NUMBER, AND THE RECEPTACLE PIN NUMBER SEPARATED BY A PERIOD (E.G. 02.30 INDICATES LOCATION 2, PIN 30). OTHER CONNECTIONS SUCH AS TO A TERMINAL BOARD ARE INDICATED BY THE DEVICE NOMENCLATURE AND A POINT NUMBER (E.G. TB.01). A WIRE BETWEEN TWO POINTS IS INDICATED BY A DASH BETWEEN THE TWO CONNECTION POINT NOMENCLATURES (E.G. TB.21 - 23.06). A CONNECTION MADE BY BUS BAR AS SHOWN IN SKETCH "B" IS INDICATED IN THE TABLE AT THE BEGINNING OF THE WIRE TABLE.

3. THE RECEPTACLE IS NUMBERED CORRESPONDING TO ITS INDIVIDUAL PINS. CARDS HAVE TAB NUMBERS WHICH CORRESPOND TO PIN NUMBERS ON THE RECEPTACLE. FOR A CARD WITH TABS ON BOTH SIDES THE CORRESPONDING RECEPTACLE HAS 64 INDIVIDUAL PINS. A CARD WITH TABS ON ONE SIDE HAS A CORRESPONDING RECEPTACLE WITH 32 PINS.

4. ON PRINTED CIRCUIT CAR USED IN THIS REGULATOR THE LETTERS "AA" AFTER THE BASIC CARD CATALOG NUMBER IN DICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBERS AND GRO NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS AB, AC, AD, ETC., ARE DIRECTLY IN.ERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARD. AN EARLIER DESIGN CANNOT REPLACE A LATER DESIGN, E.G. AN AA CANNOT REPLACE AN "AB".

5 LOCATION NUMBER 24 HAS NO PERMANENT CARD, IT IS A TEST RECEPTACLE AND MAY BE USED WITH THE TEST CARD IN LOCATION NUMBER 23

DRIVER SUBASSEMBLY EXTERNAL CONNECTIONS



601—FOR AF-3061 INVERTER ASSEMBLIES
602—FOR AF-3060 INVERTER ASSEMBLIES

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV NO	REVISIONS	DATE	APPROVED
0			

No revisions are to be made to this drawing without approval of the Development Engineering Section of the Speed Variator Department

APPLIED PRACTICES 718A505 — UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
SURFACES √
TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ±, DECIMALS ±, ANGLES ±

FIRST MADE FOR STANDARD LINE 33IX358AAG01,G02

TITLE CONNECTION DIAGRAM AF-3062 DRIVER

GENERAL ELECTRIC — SPEED VARIATOR — ERIE, PA

DO NOT SCALE LOCATION

36C764839AA REV NO 2 — CONT ON SHEET 3

PRINTS TO: JA(CD) 51(S) | 1T1BW 58(3)23 | 5L(BW)5 50(BK) | 50c(B) | 50c(BK) | 5T3950

36C764839AA REV NO 2 — CONT ON SHEET 3

36C764839AA REV NO 2 — CONT ON SHEET 3

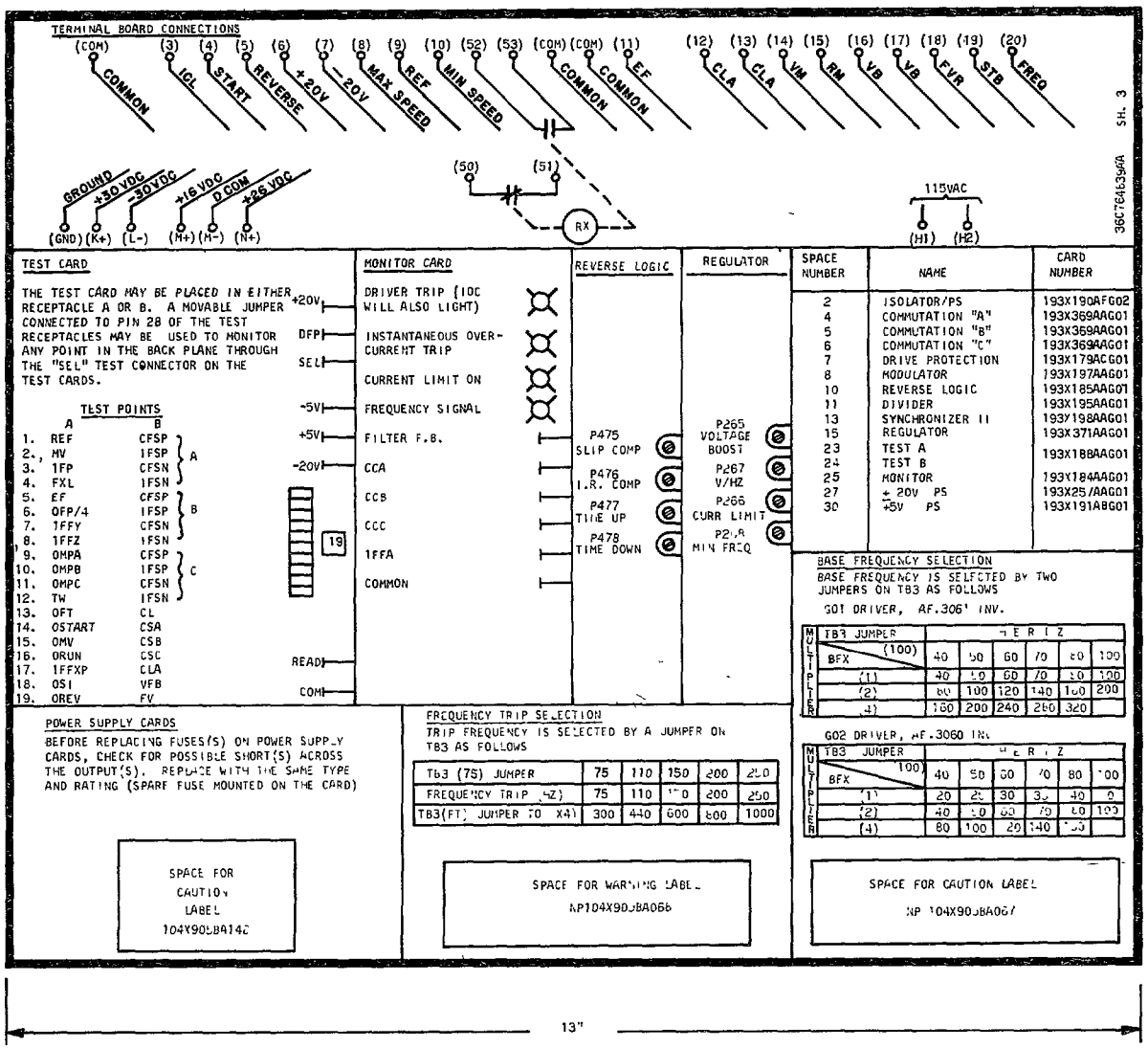
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

TITLE LABEL ARTWORK & NOTES
AF 3062 DRIVER

FIRST MADE FOR STANDARD LINE

331X358AAG01, G02



SPACE NUMBER	NAME	CARD NUMBER
2	ISOLATOR/PS	193X190AFG02
4	COMMUTATION "A"	193X369AAG01
5	COMMUTATION "B"	193X369AAG01
6	COMMUTATION "C"	193X369AAG01
7	DRIVE PROTECTION	193X179ACG01
8	MODULATOR	193X197AAG01
10	REVERSE LOGIC	193X185AAG01
11	DIVIDER	193X195AAG01
13	SYNCHRONIZER II	193Y198AAG01
15	REGULATOR	193X371AAG01
23	TEST A	193X188AAG01
24	TEST B	
25	MONITOR	193X184AAG01
27	± 20V PS	193X257AAG01
30	+5V PS	193X191ABG01

POWER SUPPLY CARDS

BEFORE REPLACING FUSES(S) ON POWER SUPPLY CARDS, CHECK FOR POSSIBLE SHORT(S) ACROSS THE OUTPUT(S). REPLACE WITH THE SAME TYPE AND RATING (SPARE FUSE MOUNTED ON THE CARD)

SPACE FOR CAUTION LABEL 104X90DBA140

FREQUENCY TRIP SELECTION

TRIP FREQUENCY IS SELECTED BY A JUMPER ON TB3 AS FOLLOWS

TB3 (75) JUMPER	75	110	150	200	250
FREQUENCY TRIP (HZ)	75	110	150	200	250
TB3(FT) JUMPER TO X4)	300	440	600	800	1000

SPACE FOR WARNING LABEL NP104X90DBA066

BASE FREQUENCY SELECTION

BASE FREQUENCY IS SELECTED BY TWO JUMPERS ON TB3 AS FOLLOWS

G01 DRIVER, AF.306' INV.

MULTIPLIER: TB3 JUMPER \ BFX	HERTZ (100)	40	50	60	70	80	100
(1)		40	50	60	70	80	100
(2)		80	100	120	140	160	200
(4)		160	200	240	280	320	

G02 DRIVER, AF.3060 INV.

MULTIPLIER: TB3 JUMPER \ BFX	HERTZ (100)	40	50	60	70	80	100
(1)		20	25	30	35	40	50
(2)		40	50	60	70	80	100
(4)		80	100	120	140	160	

SPACE FOR CAUTION LABEL NP 104X90DBA067

DRIVER DOOR LABEL

1. SIZE AS INDICATED.
2. PRINT ON FASSON WHITE CONVENTION BADGE (LAPEL) STOCK NO. 65P.
3. CUT TO SIZE.

11"

13"

FOLD

CONT ON SHEET FILM NO REV NO 0

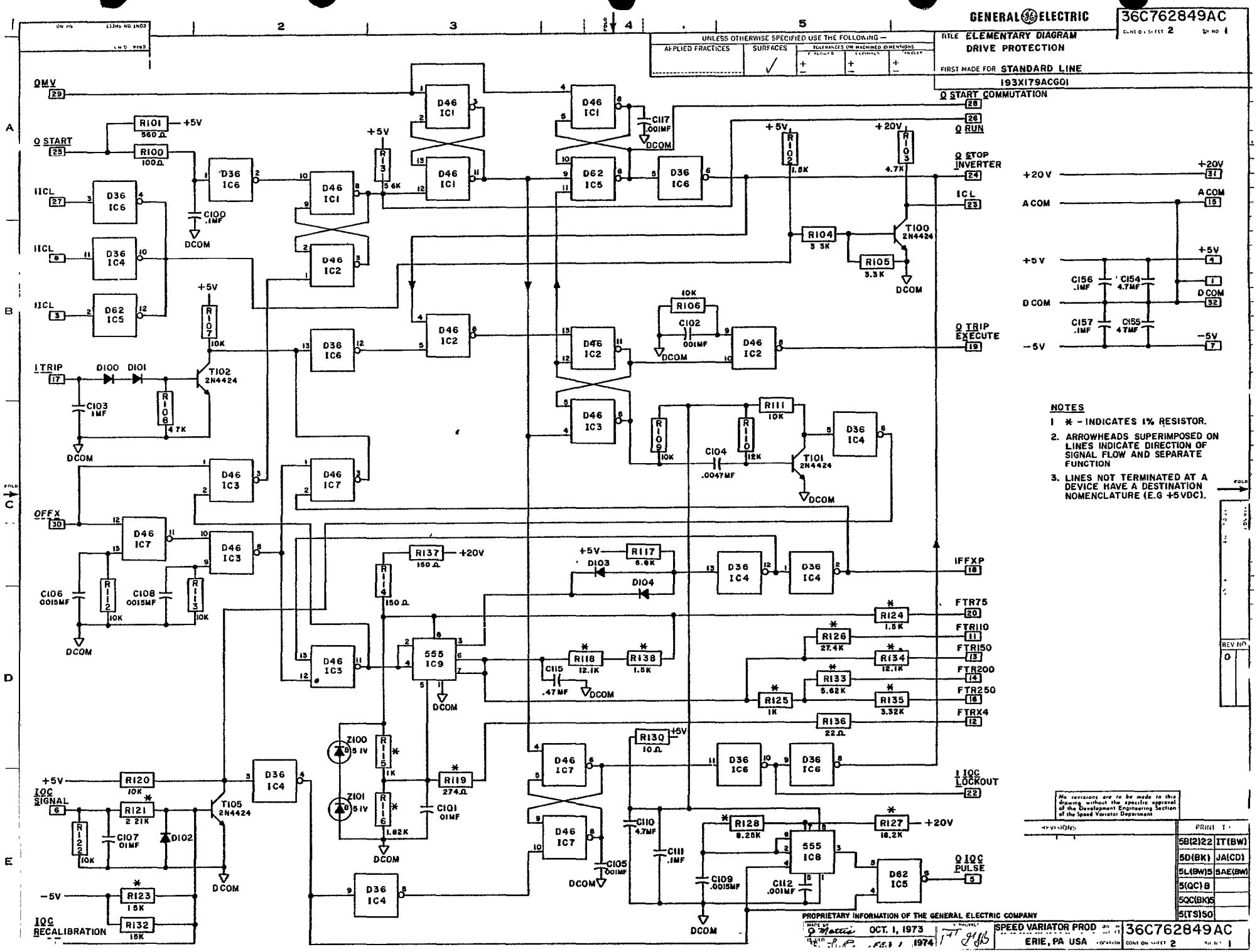
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REVISIONS

PRINTS TO

5(S)	JA(KD)
5B(3)23	IT(BW)
5D(BK)	5L(BW)5
5QC(B)	
5QC(BK)5	
5TS)50	

MADE BY W. Wolszczak MAR 12, 1974
ISSUED
APPROVED
SPEED VARIATOR ERIE, PA.
DIV OR DEPT LOCATION
36C764839AA
CONT ON SHEET FL SH NO 3


GENERAL ELECTRIC
36C762849AC
TITLE ELEMENTARY DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179ACG01
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED FRACTICES
SURFACES
Q START COMMUTATION
Q RUN
Q STOP INVERTER
ICL
Q TRIP EXECUTE
IFFXP
FTR75
FTR110
FTR150
FTR200
FTR250
FTRX4
I IOC LOCKOUT
Q IOC PULSE
OMV
Q START
IICL
ITRIP
OFFX
IOC SIGNAL
IOC RECALIBRATION
+5V
+20V
-5V
DCOM
ACOM
555 IC9
555 IC8
NOTES
1 * - INDICATES 1% RESISTOR.
2. ARROWHEADS SUPERIMPOSED ON LINES INDICATE DIRECTION OF SIGNAL FLOW AND SEPARATE FUNCTION
3. LINES NOT TERMINATED AT A DEVICE HAVE A DESTINATION NOMENCLATURE (E.G +5VDC).
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR PROD
ERIE, PA USA
36C762849AC

GENERAL ELECTRIC

36C762849AC

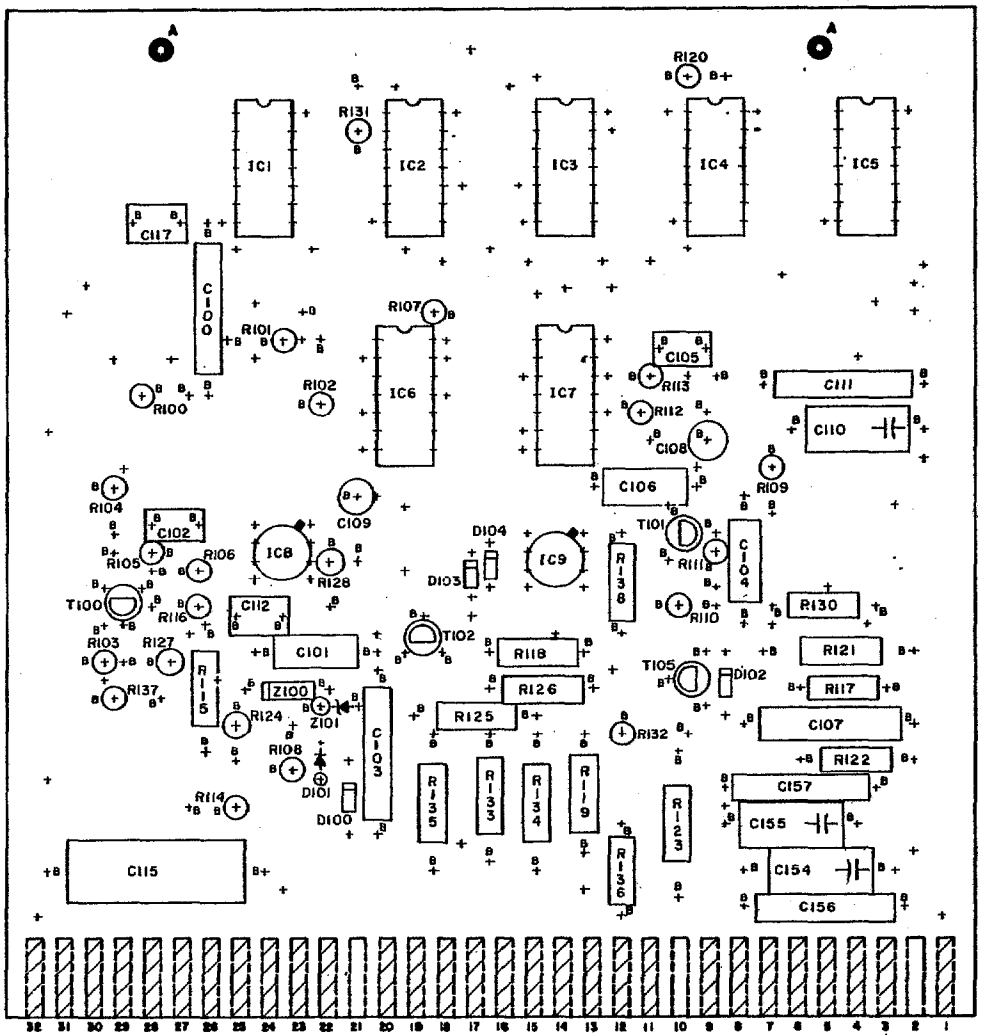
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES: 710A816 | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179AG001

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

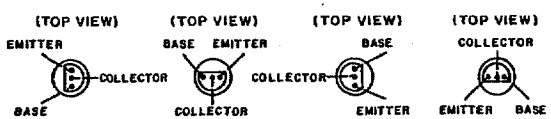
LOC.	DIA.	QUAN.
A	.157	2
B	.040	130

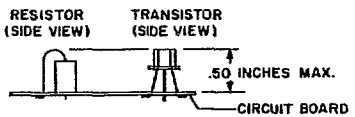


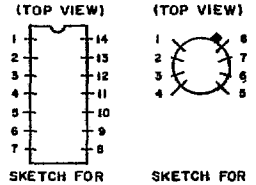
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $6.150^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

TRANSISTOR LEAD SKETCHES







2X SCALE	36A352983CA005 REAR ETCHED BOARD DWG.	36A352982CA005 FRONT ETCHED BOARD DWG.	1 FRONT SPACE UNIT REQUIREMENT	0 BACK SPACE UNIT REQUIREMENT

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varistor Department

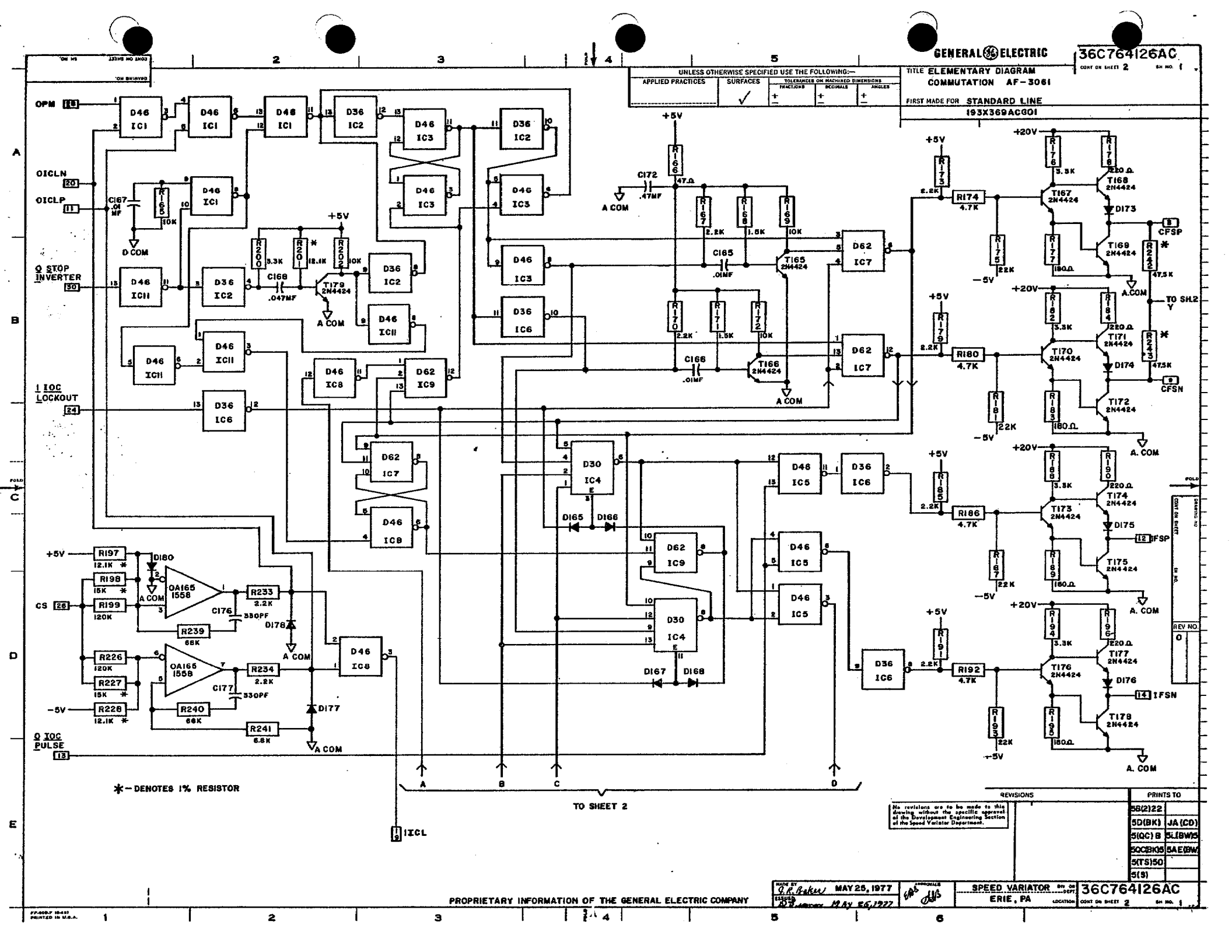
REV NO. 0

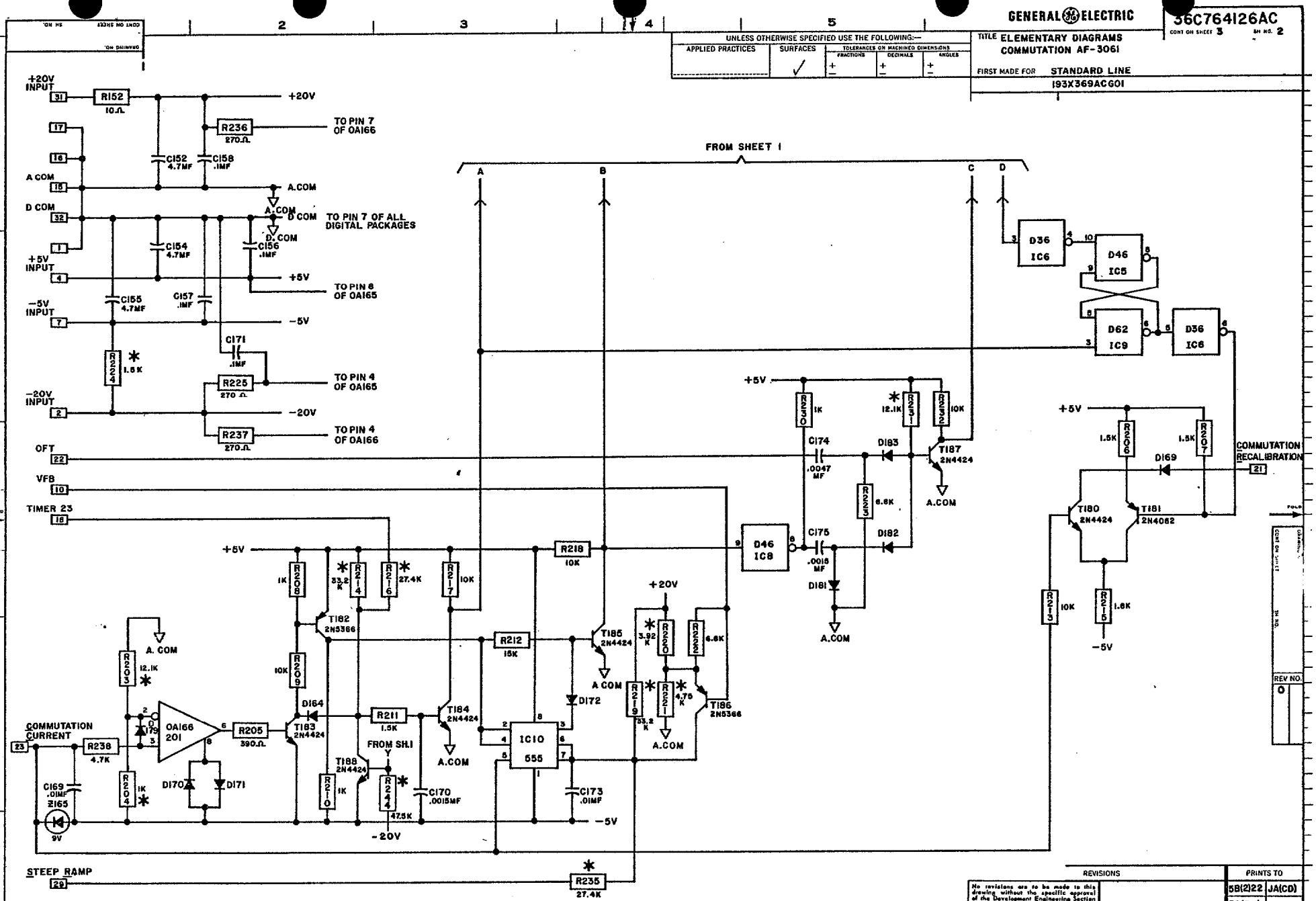
PRINTS TO

5QC(B)	5 (B)
5B(2)22	5D(BK)
5L(BW)5	5D(B)
5QC(BK)5	5AE(BW)
5(TS)50	J4(CD)
1T(BW)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

ISSUED BY G. Mastis OCT. 3, 1973	APPROVED	SPEED VARIATOR PROD DIV OR DEPT	36C762849AC	
ISSUED E. Sedgwick FEB 1, 1974		ERIE, PA. USA LOCATION	CONT ON SHEET FL	SH NO 2


GENERAL ELECTRIC
36C764126AC
TITLE ELEMENTARY DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
OPM 18
OICLN 20
OICLP 11
Q STOP INVERTER 30
I IOC LOCKOUT 24
Q IOC PULSE 13
CS 26
D COM
A COM
+5V
-5V
+20V
TO SHEET 2
IICL
*—DENOTES 1% RESISTOR
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
SPEED VARIATOR
ERIE, PA
MADE BY G.R. Baker MAY 25, 1977
ISSUED MAY 26, 1977
36C764126AC


+20V INPUT
R152 10Ω
+20V
R236 270Ω
TO PIN 7 OF OA166
C152 4.7MF
C158 .1MF
A COM
A.COM
D COM
TO PIN 7 OF ALL DIGITAL PACKAGES
C154 4.7MF
C156 .1MF
+5V INPUT
+5V
TO PIN 8 OF OA165
C155 4.7MF
C157 .1MF
-5V INPUT
-5V
C171 .1MF
R224 1.5K
R225 270Ω
TO PIN 4 OF OA165
-20V INPUT
-20V
R237 270Ω
TO PIN 4 OF OA166
OFT
VFB
TIMER 23
FROM SHEET 1
D36 IC6
D46 IC5
D62 IC9
D36 IC6
+5V
R250 1K
R251 12.1K
R252 10K
C174 .0047 MF
D183
T187 2N4424
A.COM
R253 6.8K
C175 .0018 MF
D182
D181
A.COM
D46 IC8
R218 10K
R208 1K
R214 33.2K
R216 27.4K
R217 10K
R209 10K
T182 2N5386
R212 15K
T185 2N4424
A COM
D172
+20V
R220 3.92K
R222 4.75K
R221 6.8K
R219 33.2K
T186 2N5366
A.COM
R206 1.5K
R207 1.5K
D169
COMMUTATION RECALIBRATION
21
T180 2N4424
T181 2N4062
R213 10K
R215 1.8K
-5V
A. COM
R203 12.1K
COMMUTATION CURRENT
23
R238 4.7K
OA166 201
R205 390Ω
D164
T183 2N4424
R211 1.5K
T184 2N4424
A.COM
FROM SH.1
T188 2N4424
R210 1K
R204 1K
D170
D171
C169 .01MF 2165
9V
C170 .0015MF
-20V
IC10 555
C173 .01MF
-5V
STEEP RAMP
29
R235 27.4K
REV NO. 0
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
5B(2)22 JA(CD)
5D(BK)
5(QC)B 5L(BW)5
5QC(BK)5 5AE(BW)
5(TS)5D
5(S)
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
MADE BY R.A. Baker MAY 25, 1977
ISSUED MAY 25, 1977
APPROVALS
speed variator DIV OR DEPT
ERIE, PA. LOCATION
36C764126AC CONT ON SHEET 3 SH NO. 2
GENERAL ELECTRIC
36C764126AC CONT ON SHEET 3 SH NO. 2
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES
TITLE ELEMENTARY DIAGRAMS COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01

GENERAL ELECTRIC

36C764126AC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

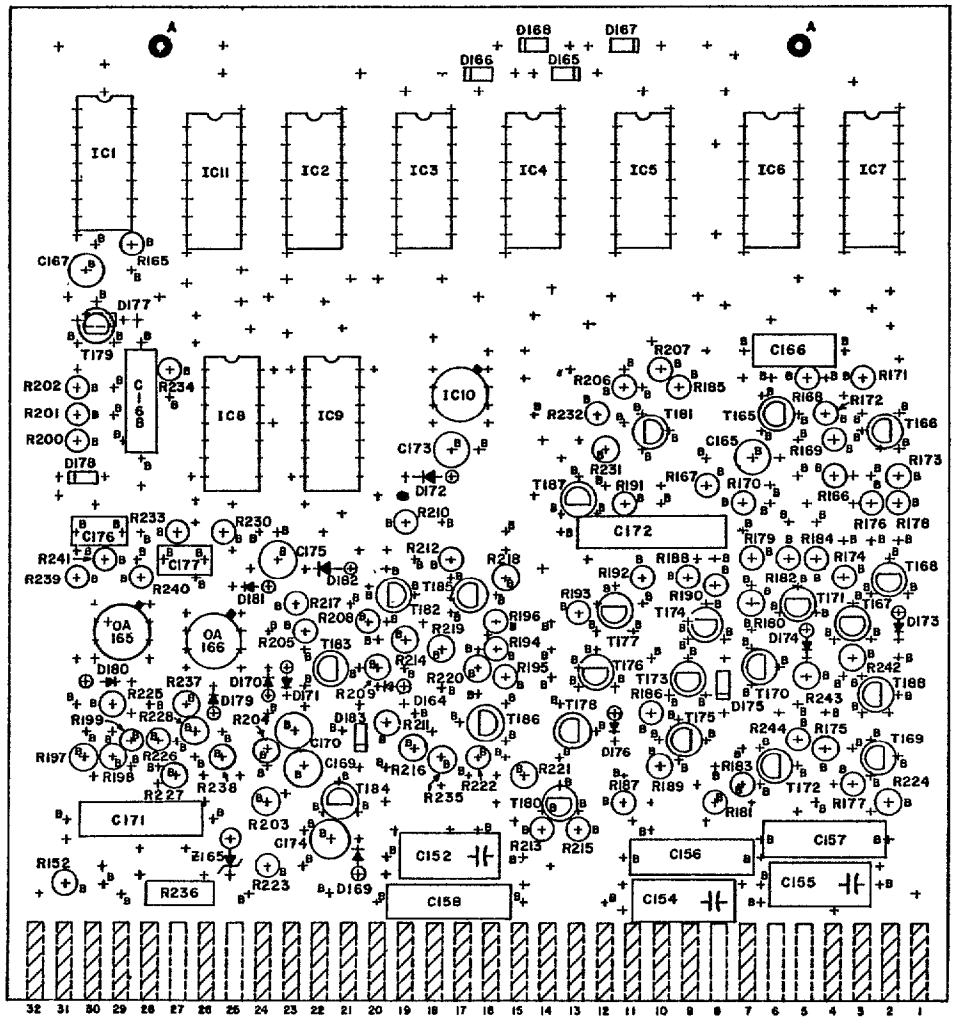
TITLE PRINTED CIRCUIT DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369AC601

CONT ON SHEET FL. SH NO 3

HOLE TABULATION

ALL HOLES 032 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	157	2
B	.040	269

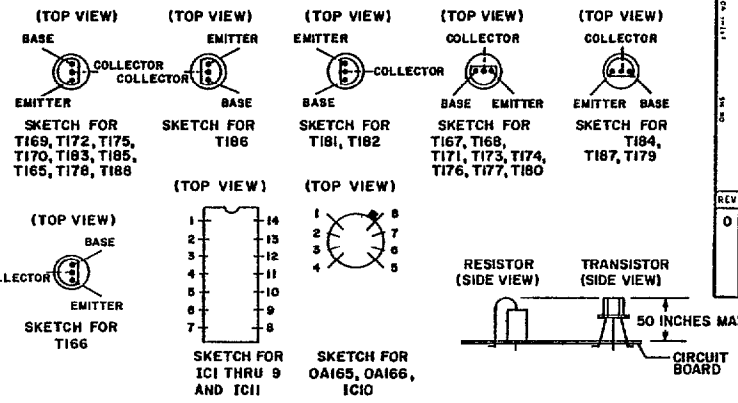


SEE NOTE 1 & 2

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015} \times 5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES



REV NO 0

2X SCALE	36A353788AB005 REAR ETCHED BOARD DWG.	36A353789AB005 FRONT ETCHED BOARD DWG.	1 FRONT SPACE UNIT REQUIREMENT	0 BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

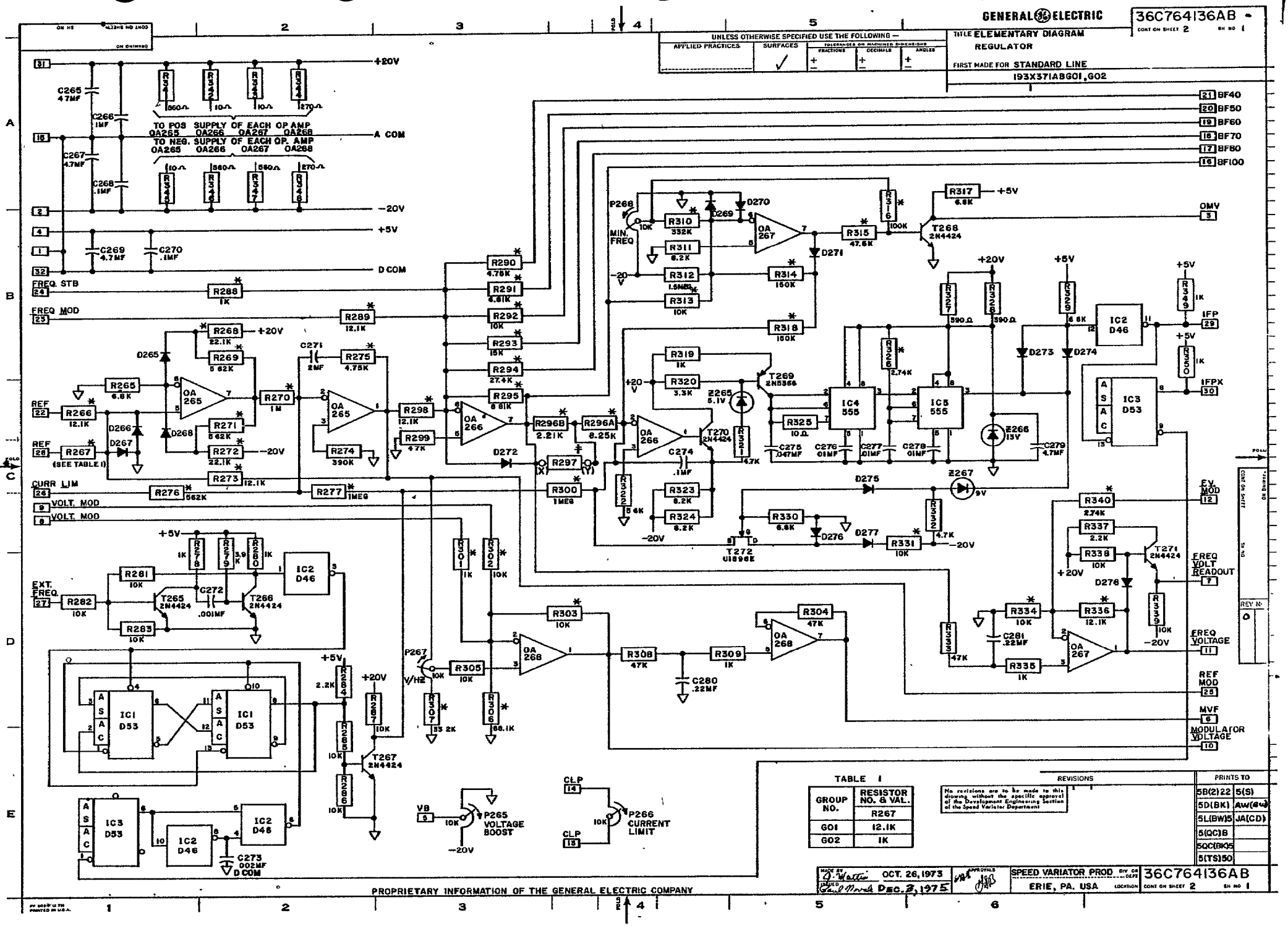
5B(2)22	
5D(BK)	JA(CD)
5(QC)B	5L(BW)5
50C(BK)5	5AE(BW)
5(TS)50	
5(S)	

MADE BY D.R. Baker MAY 25, 1977
ISSUED Approved May 25, 1977
APPROVALS ENGR. MGR.

SPEED VARIATOR PROD. DEPT. OR DEPT. ERIE, PA. LOCATION

36C764126AC CONT ON SHEET FL. SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764136AB
ELEMENTARY DIAGRAM
REGULATOR
FIRST MADE FOR STANDARD LINE
193X371ABG01, G02
+20V
-20V
+5V
D COM
A COM
TO POS SUPPLY OF EACH OP AMP
OA265 OA266 OA267 OA268
TO NEG. SUPPLY OF EACH OP. AMP
OA265 OA266 OA267 OA268
FREQ. STB
FREQ MOD
REF
CURR LIM
VOLT. MOD
EXT. FREQ
MIN. FREQ
V/Hz
VB
VOLTAGE BOOST
CLP
CURRENT LIMIT
IC1 D53
IC2 D46
IC3 D53
IC4 555
IC5 555
OA 265
OA 266
OA 267
OA 268
T265 2N4424
T266 2N4424
T267 2N4424
T268 2N4424
T269 2N5366
T270 2N4424
T271 2N4424
T272 U1898E
BF40
BF50
BF60
BF70
BF80
BF100
OMV
IFP
IFPX
FV MOD
FREQ VOLT READOUT
FREQ VOLTAGE
REF MOD
MVF
MODULATOR VOLTAGE
TABLE I
GROUP NO.
RESISTOR NO. & VAL.
R267
G01 12.1K
G02 1K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department
PRINTS TO
5B(2)22 5(S)
5D(BK) AW(GW)
5L(BW)5 JA(CD)
5(QC)B
5QC(BK)5
5(TS)50
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
OCT. 26, 1973
Dec. 3, 1975
SPEED VARIATOR PROD
ERIE, PA. USA
36C764136AB

GENERAL ELECTRIC

36C764136AB CONT ON SHEET FL. SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

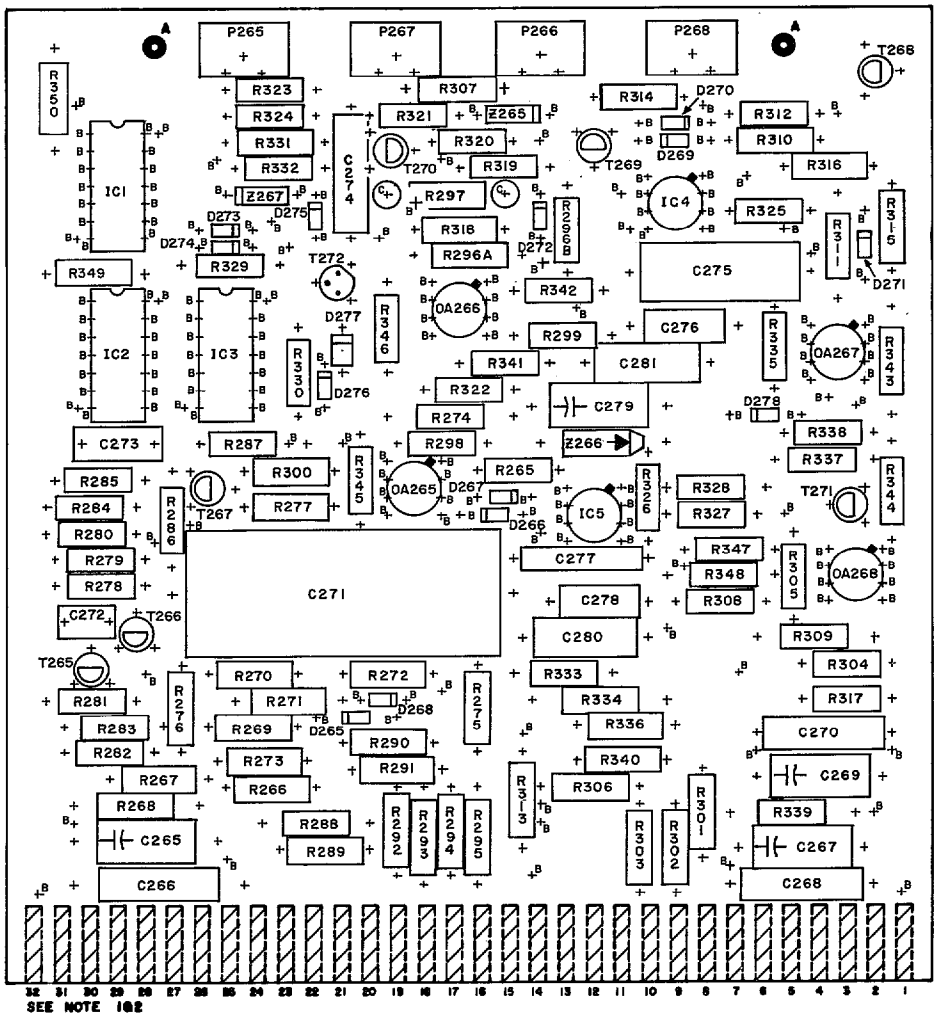
APPLIED PRACTICES	SURFACES	TOLERANCES OR MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
....ZIOASIS.....	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM

REGULATOR CARD

FIRST MADE FOR STANDARD LINE AF-3061

193X371ABG01,G02



HOLE TABULATION

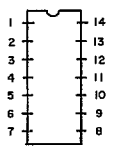
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.032	168
C	.081	2

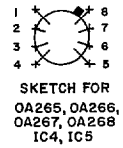
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND OP AMPS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.
5. FOR-G02 ONLY STAMP ADHESIVE LABEL-G02. ADD LABEL AS SUFFIX TO 193X371AA IN PLACE OF G01 STAMPED ON PRINTED CIRCUIT BOARD.

IC & OA LEAD SKETCHES

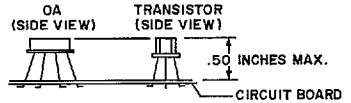


SKETCH FOR IC1, IC2, IC3

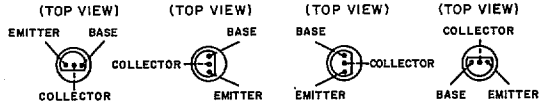


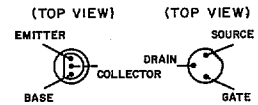
SKETCH FOR OA265, OA266, OA267, OA268 IC4, IC5

COMPONENT HEIGHT SKETCH



TRANSISTOR LEAD SKETCHES





2X	36A353697AC005	36A353698AC005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1 M. Cotty Nov. 30, 1978 PER SV-883 # 37

REV NO. 1

PRINTS TO

5B(B)M	5J(BW)
5R(2)BW	TS(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

DRAWN BY J. Mattia NOV. 21, 1973

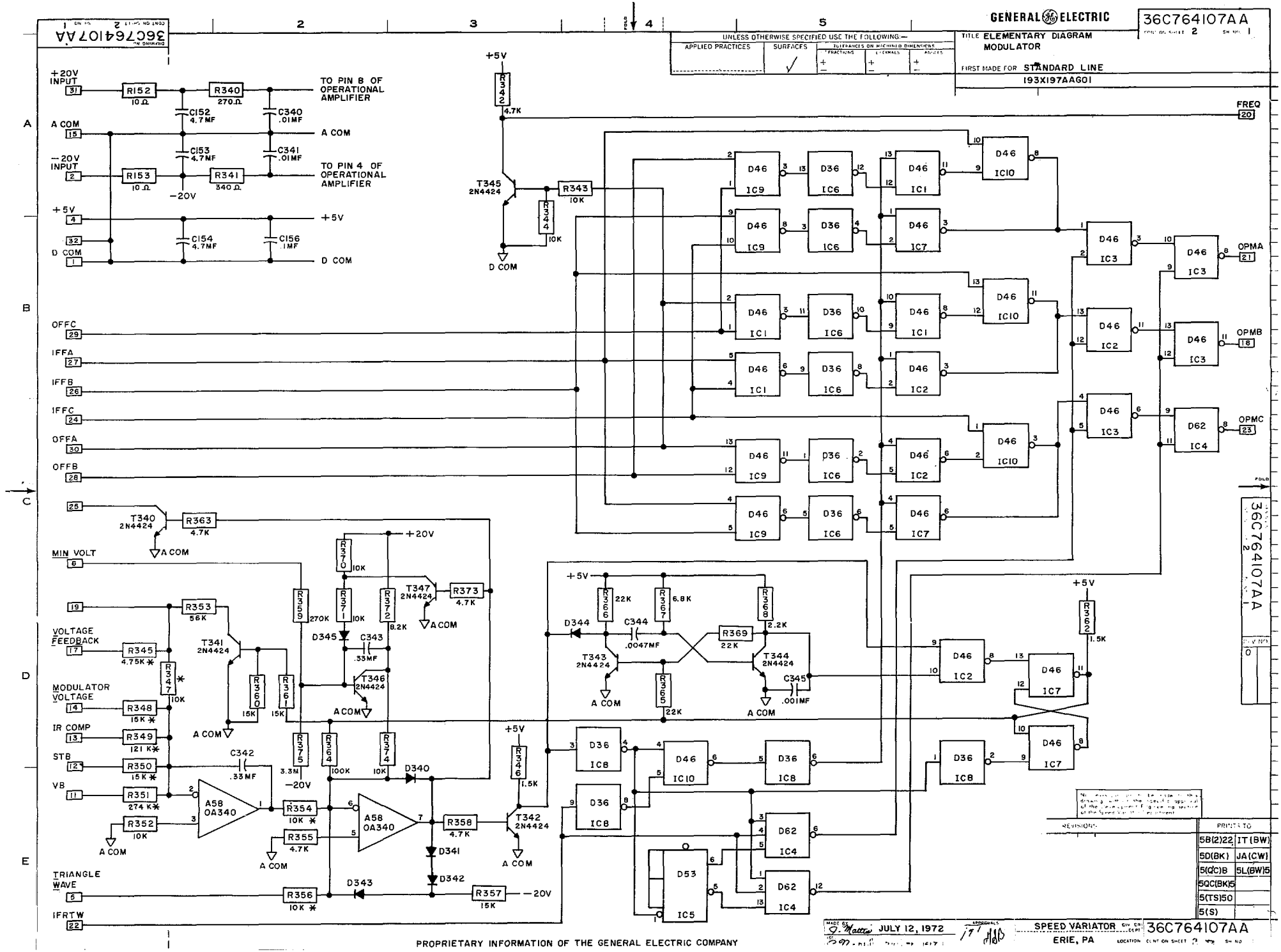
TRACED M. Cotty Nov. 30, 1978

APPROVALS ERS JJB

SPEED VARIATOR PROD. DEPT ERIE, PA. USA LOCATION

36C764136AB CONT ON SHEET FL. SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764107AA
TITLE ELEMENTARY DIAGRAM MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
+20V INPUT
-20V INPUT
A COM
D COM
+5V
TO PIN 8 OF OPERATIONAL AMPLIFIER
TO PIN 4 OF OPERATIONAL AMPLIFIER
R152 10Ω
R153 10Ω
R340 270Ω
R341 340Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C156 .1MF
C340 .01MF
C341 .01MF
FREQ
OFFC
IFFA
IFFB
IFFC
OFFA
OFFB
MIN VOLT
VOLTAGE FEEDBACK
MODULATOR VOLTAGE
IR COMP
STB
VB
TRIANGLE WAVE
IFRTW
OPMA
OPMB
OPMC
T340 2N4424
T341 2N4424
T342 2N4424
T343 2N4424
T344 2N4424
T345 2N4424
T346 2N4424
T347 2N4424
R343 10K
R344 10K
R345 4.75K*
R348 15K*
R349 121K*
R350 15K*
R351 274K*
R352 10K
R353 56K
R354 10K*
R355 4.7K
R356 10K*
R357 15K
R358 4.7K
R359 270K
R360 15K
R361 15K
R362 1.5K
R363 4.7K
R364 100K
R365 22K
R366 22K
R367 6.8K
R368 2.2K
R369 22K
R370 10K
R371 10K
R372 8.2K
R373 4.7K
R374 10K
R346 1.5K
R347 10K*
C342 .33MF
C343 .33MF
C344 .0047MF
C345 .001MF
D340
D341
D342
D343
D344
D345
A58 0A340
D46 IC1
D46 IC2
D46 IC3
D46 IC7
D46 IC9
D46 IC10
D36 IC6
D36 IC8
D62 IC4
D53 IC5
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
S. Matte JULY 12, 1972
SPEED VARIATOR
ERIE, PA
36C764107AA
PRINTS TO
5B(2)22 1T(BW)
5D(BK) JA(CW)
5(QC)B 5L(BW)5
5QC(BK)5
5(TS)50
5(S)

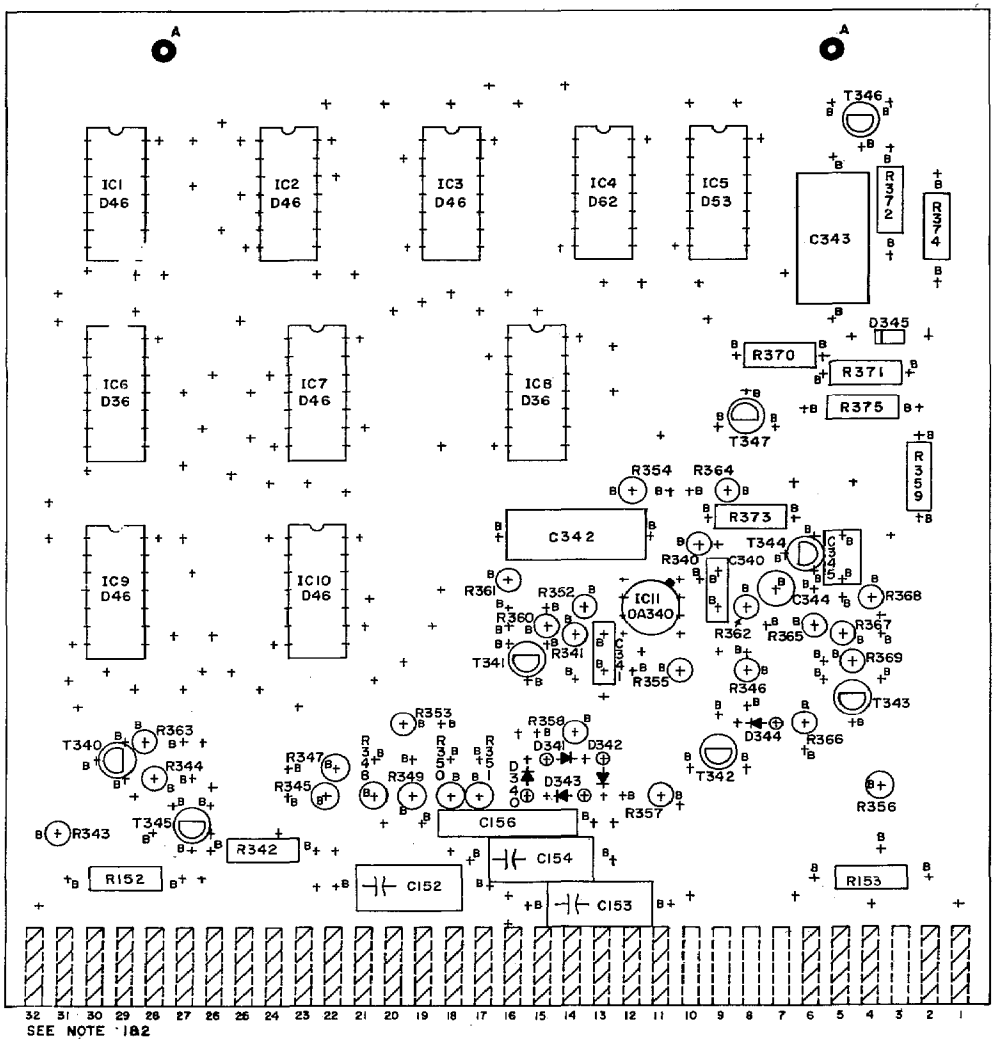
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A815	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01

HOLE TABULATION

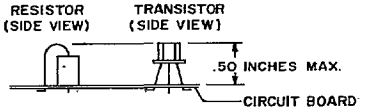
ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.040	120

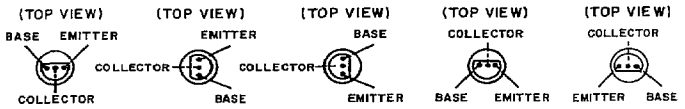


NOTES

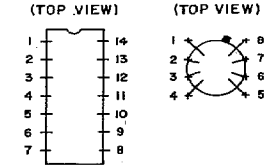
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.



TRANSISTOR LEAD SKETCHES



IC LEAD SKETCHES



2X	36A352956AA005	36A352955AA005	1	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varitor Department.

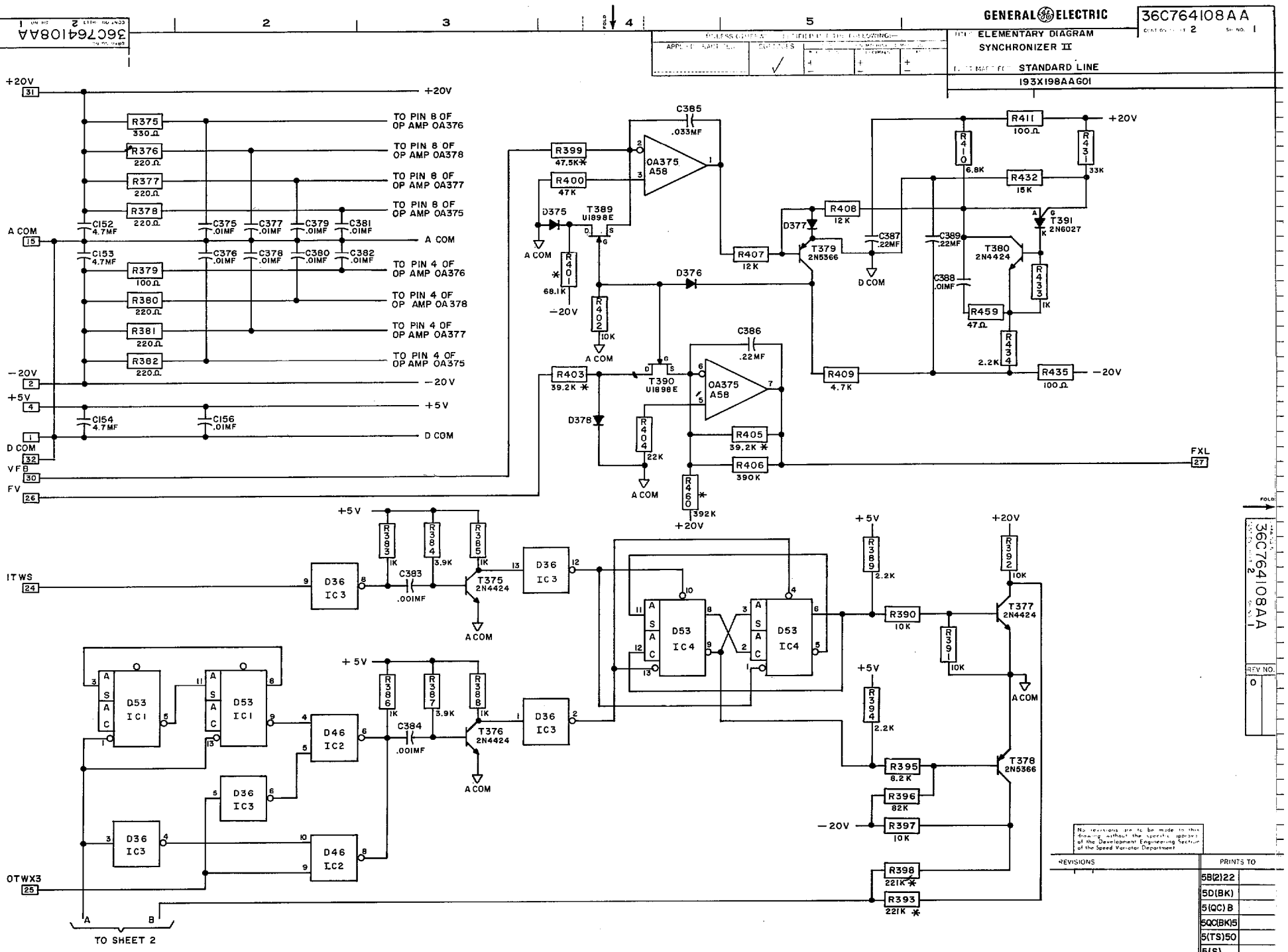
REVISIONS

PRINTS TO	
5B(2)22	IT(BW)
5D(BK)	JA(CD)
5(QC)B	5L(BW)5
5QC(BK)5	
5(TS)50	
5(S)	

REV NO.
0

36C764107AA FL SH NO 2

DRAWN BY P. Mattis AUG. 25, 1972 APPROVED SPEED VARIATOR DIV OR DEPT 36C764107AA


GENERAL ELECTRIC
36C764108AA
ELEMENTARY DIAGRAM
SYNCHRONIZER II
STANDARD LINE
193X198AAG01
+20V
A COM
D COM
VFB
FV
ITWS
OTWX3
FXL
TO SHEET 2
R375 330Ω
R376 220Ω
R377 220Ω
R378 220Ω
R379 100Ω
R380 220Ω
R381 220Ω
R382 220Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C156 .01MF
C375 .01MF
C376 .01MF
C377 .01MF
C378 .01MF
C379 .01MF
C380 .01MF
C381 .01MF
C382 .01MF
TO PIN 8 OF OP AMP OA376
TO PIN 8 OF OP AMP OA378
TO PIN 8 OF OP AMP OA377
TO PIN 8 OF OP AMP OA375
TO PIN 4 OF OP AMP OA376
TO PIN 4 OF OP AMP OA378
TO PIN 4 OF OP AMP OA377
TO PIN 4 OF OP AMP OA375
R399 47.5K*
R400 47K
R401 68.1K*
R402 10K
R403 39.2K*
R404 22K
R405 39.2K*
R406 390K
R407 12K
R408 12K
R409 4.7K
R410 6.8K
R411 100Ω
R431 33K
R432 15K
R433 1K
R434 2.2K
R435 100Ω
R459 47Ω
R460 392K*
C385 .033MF
C386 .22MF
C387 .22MF
C388 .01MF
C389 .22MF
OA375 A58
T389 U1898E
T390 U1898E
T379 2N5366
T380 2N4424
T391 2N6027
D375
D376
D377
D378
R383 1K
R384 3.9K
R385 1K
C383 .001MF
T375 2N4424
R386 1K
R387 3.9K
R388 1K
C384 .001MF
T376 2N4424
D36 IC3
D53 IC1
D53 IC4
D46 IC2
R389 2.2K
R390 10K
R391 10K
R392 10K
R394 2.2K
R395 8.2K
R396 82K
R397 10K
R398 221K*
R393 221K*
T377 2N4424
T378 2N5366
REVISIONS
PRINTS TO
5B(2)22
5D(BK)
5(QC)8
5QC(BK)5
5(TS)50
5(S)
JULY 12, 1972
SPEED VARIATOR
ERIE, PA.
36C764108AA

GENERAL ELECTRIC 36C764108AA

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES. TOLERANCES: FRACTIONS ±1/64, DECIMALS ±.005, ANGLES ±1°. APPLIED FRACTILES: SURFACES √

TITLE: ELEMENTARY DIAGRAM SYNCHRONIZER II

SIZE 3 SHEET 2

FOR STANDARD LINE 193X198AAG01

FROM SHEET 1

D379 D380 D381 D382 D383 D384 D385 D386

C390 .22MF · C391 .01MF · C392 .001MF · C393 4.7MF · C394 .01MF · C395 .1MF

R412 27K · R413 100K · R414 10K * · R415 4.7K · R416 10K * · R417 6.81K * · R418 4.7K · R419 150K · R420 15K * · R421 10K · R422 10K · R423 3.3K · R424 100K · R425 33K · R426 5.6K · R427 100K · R428 2.2K · R429 15K · R430 100K · R436 3.3K · R437 2.2K · R438 560Ω · R439 560Ω · R441 4.75K * · R442 18.2K * · R443 4.7K · R444 4.75K * · R445 18.2K * · R446 4.7K · R447 4.75K * · R448 18.2K * · R449 4.7K · R450 22Ω · R451 2.21K * · R452 6.81K * · R453 6.81K * · R454 10K * · R455 475Ω · R456 1K * · R457 1K * · R458 332Ω

T381 2N4424 · T382 2N5366 · T383 2N4424 · T384 2N4424 · T385 2N5366 · T386 2N5366 · T387 2N4424 · T388 U1898E

OA376 A58 · OA377 A58 · OA378 A58 · OA376 A58 · OA377 A5A · OA378 A58

+20V · +5V · −20V · −20V · +20V · +5V · +20V · −20V

A COM

TRIANGLE WAVE 18

CFA 23 · CFB 20 · CFC 14

CSA 22 · CSB 18 · CSC 12

OFT 9

5B(2)22	
5D(BK)	
5(QC)8	
5QC(BK)5	
5(TS)50	
5(S)	

DRAWN BY D. Mazza JULY 11, 1972 CHECKED BY [illegible] SPEED VARIATOR ENG. OR DEPT. 36C764108AA

36C764108AA SHT 3 REV 2

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

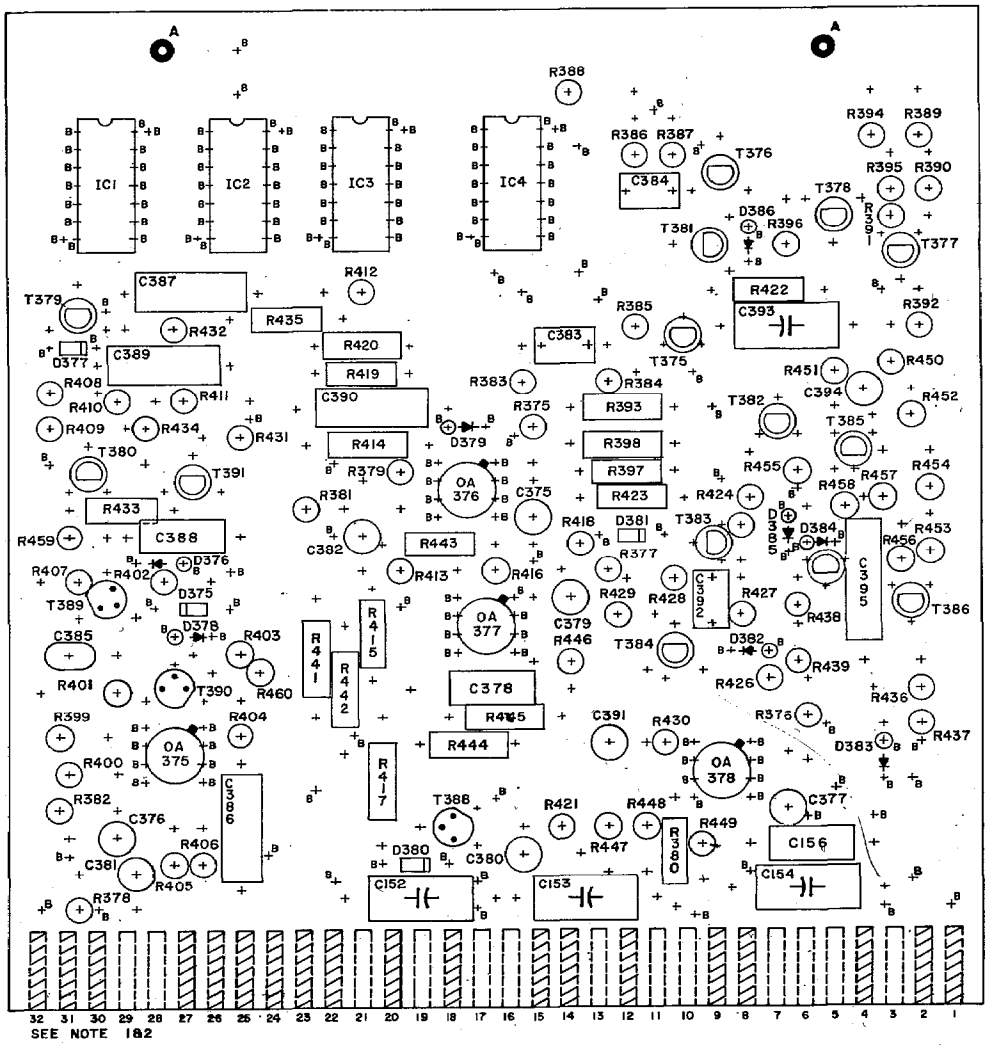
TITLE PRINTED CIRCUIT DIAGRAM
SYNCHRONIZER II
FIRST MADE FOR STANDARD LINE
193X198AAG01

36C764108AA CONT ON SHEET FL SH NO 3

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

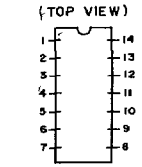
LOC.	DIA.	QUAN.
A	.157	2
B	.032	164



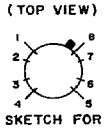
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

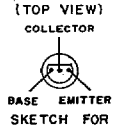
TRANSISTOR SKETCHES



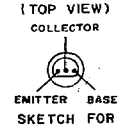
SKETCH FOR IC PACKAGES



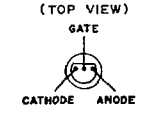
SKETCH FOR 0A375, 0A376, 0A377, 0A378



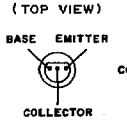
SKETCH FOR T375, T376, T379, T382, T384



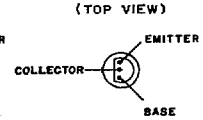
SKETCH FOR T380



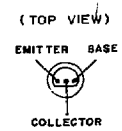
SKETCH FOR T391



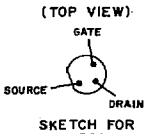
SKETCH FOR T377, T378, T386



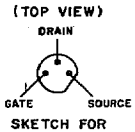
SKETCH FOR T381, T383



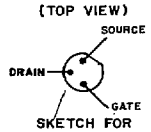
SKETCH FOR T385, T387



SKETCH FOR T389



SKETCH FOR T390



SKETCH FOR T388

2X	36A353762AA005	36A353763AA005	1	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT	BACK REQUIREMENT

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

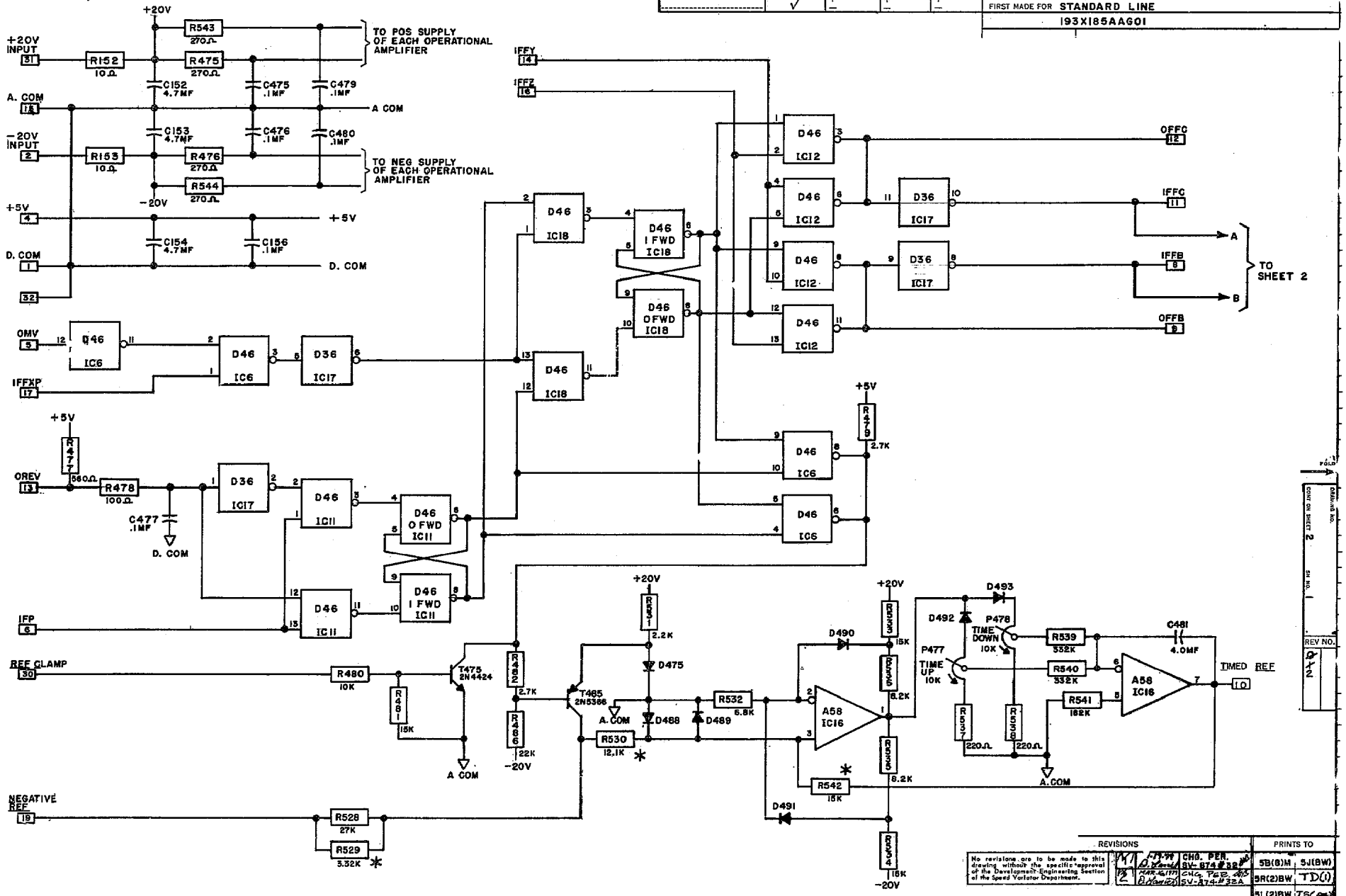
REV NO.
0

PRINTS TO

5QC(S) 5B (S)
5D(BK)
5B (2)
5QC(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY J.E. Baker ISSUED OCT. 24, 1972 — APPROVALS — SPEED VARIATOR ERIE — 36C764108AA LOCATION CONT ON SHEET FL SH NO 3



GENERAL ELECTRIC

36C762855AB

CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC

FIRST MADE FOR STANDARD LINE
193X185AAG01

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

M1	6-17-99	CHG. PER.	SV-874#32B
P3	7-28-99	CHG. PER.	SV-874#32A

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	TD()
5L(2)BW	TSC()
AW(BW)	
5D(CD)	
JA(CD)	

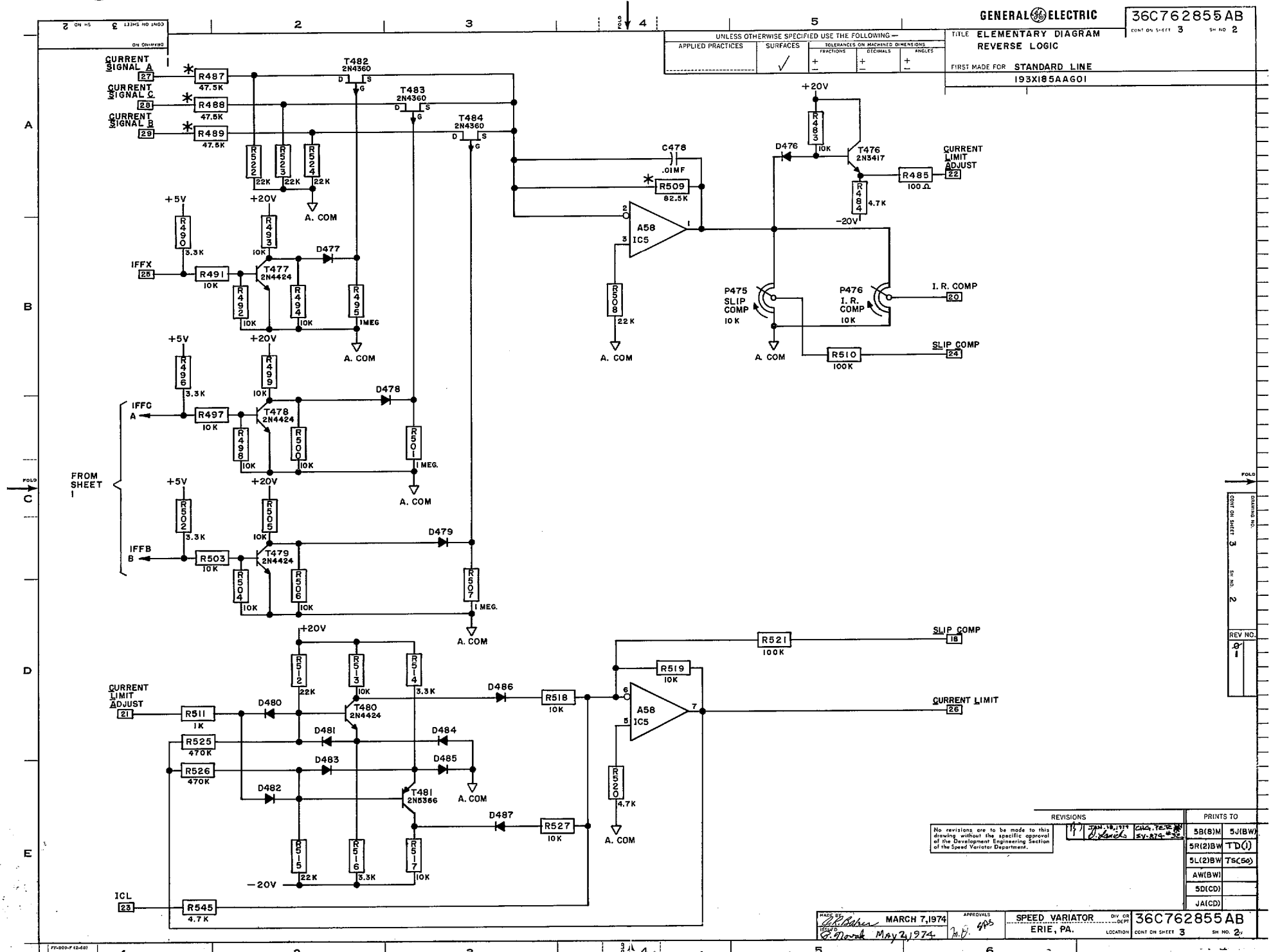
MADE BY E.C. Baker MARCH 7, 1974
CHKD BY S. Novak May 2, 1974
APPROVALS J.O. GAL

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C762855AB
CONT ON SHEET 2 SH NO 1

DRAWING NO. 36C762855AB
CONT ON SHEET 2 SH NO 1
REV NO. P

FF-BOP-F (2-68) PRINTED IN U.S.A.


GENERAL ELECTRIC
36C762855AB
TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC
FIRST MADE FOR STANDARD LINE
193X185AAG01
CURRENT SIGNAL A
CURRENT SIGNAL C
CURRENT SIGNAL B
R487 47.5K
R488 47.5K
R489 47.5K
T482 2N4360
T483 2N4360
T484 2N4360
R522 22K
R523 22K
R524 22K
A. COM
+5V
+20V
R490 3.3K
R493 10K
IFFX
R491 10K
T477 2N4424
D477
R492 10K
R494 10K
R495 1MEG
R496 3.3K
R499 10K
IFFC A
R497 10K
T478 2N4424
D478
R498 10K
R500 10K
R501 1MEG.
FROM SHEET 1
R502 3.3K
R505 10K
IFFB B
R503 10K
T479 2N4424
D479
R504 10K
R506 10K
R507 1 MEG.
C478 .01MF
R509 82.5K
A58 IC5
R508 22K
+20V
R483 10K
D476
T476 2N3417
CURRENT LIMIT ADJUST
R485 100Ω
R484 4.7K
-20V
P475 SLIP COMP 10K
P476 I.R. COMP 10K
I. R. COMP 20
R510 100K
SLIP COMP 24
R521 100K
SLIP COMP 18
R519 10K
CURRENT LIMIT 26
R520 4.7K
R512 22K
R513 10K
R514 3.3K
D486
R518 10K
CURRENT LIMIT ADJUST 21
R511 1K
D480
T480 2N4424
D481
D484
R525 470K
D483
D485
R526 470K
D482
T481 2N6356
D487
R527 10K
R515 22K
R516 3.3K
R517 10K
-20V
ICL 23
R545 4.7K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
5B(8)M 5J(BW)
5R(2)BW TD(J)
5L(2)BW TS(5Q)
AW(BW)
5D(CD)
JA(CD)
MARCH 7,1974
SPEED VARIATOR
ERIE, PA.
36C762855AB

GENERAL ELECTRIC

36C762855AB
CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
REVERSE LOGIC

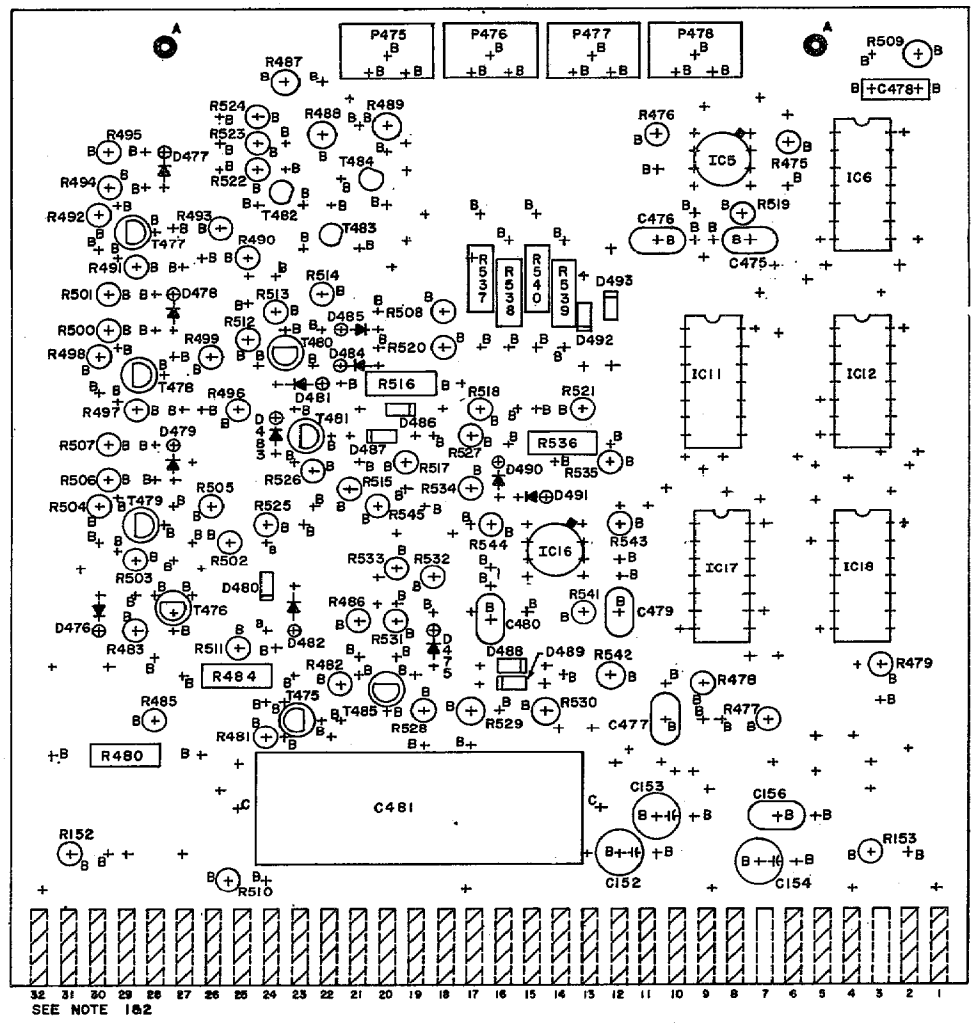
FIRST MADE FOR STANDARD LINE

193X185AAG01

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

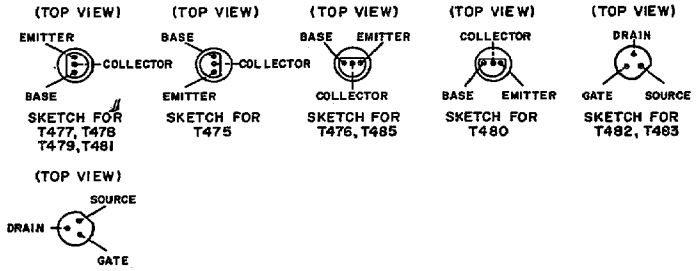
LOC.	DIA.	QUAN.
A	.157	2
B	.040	213
C	.052	2



NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES



2X	36A35297OAB005	36A35297IAB005	2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	D. Baker 11/13/1974	CHG. PER ECO 5V-874-32	
2	D. Novak 11/26/1974	CHG. PER ECO 5V-874-32A	
3	MAR. 16, 1974 D. Novak	CHG. PER ECO 5V-874-32B	

PRINTS TO

5B(8)M	5J(8W)
5R(2)BW	TD(1)
5L(2)BW	TS(2)
AW(8W)	
5D(CD)	
JA(CD)	

MADE BY E.R. Baker MARCH 8, 1974
ISSUED D. Novak MAY 2, 1974
APPROVALS M.B. EBS

SPEED VARIATOR DIV. OR DEPT
ERIE, PA. LOCATION

36C762855AB
CONT ON SHEET FL SH NO 3

SCHEMATIC DIAGRAM

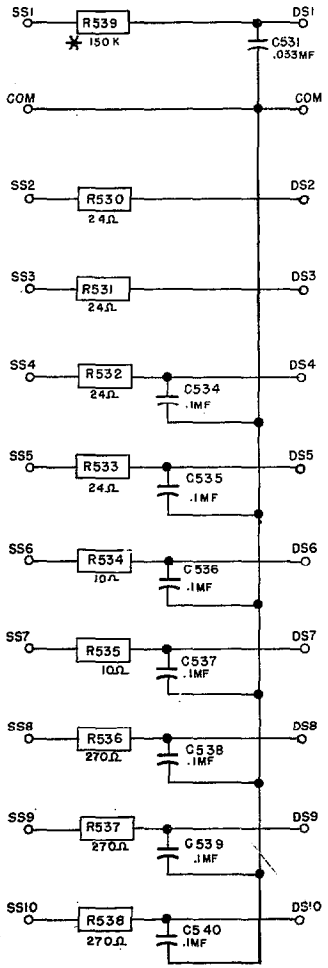
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	FRACTION	
718A515	✓		

GENERAL ELECTRIC

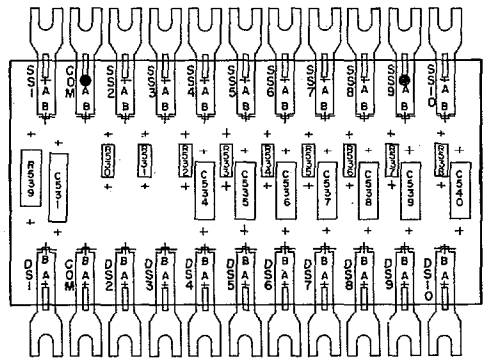
36C762868AA

PRINTED CIRCUIT DIAGRAM
SIGNAL FILTER
STANDARD LINE
193X189AAG01



✱ — INDICATES 1% RESISTANCES TOLERANCE

FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA.	QUAN.
A	.157	22
B	.101	22

NOTE: 1. CARD SIZE, $2.85^{\pm.020}$ X $5.130^{+.002}_{-.008}$

36C762868AA

Ø 1

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	942B522AA004	———	———

REVISIONS

1	J. Schmitz Nov 22, 1971	KJR	5QC(BK)5 5 (S) 5D(BK) 5(PCM) 50 5B(2) 5(TS)50 5(QC)B

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

APPROVALS

J. Schmitz June 2, 1970 EAS
DES. ENGR. JAN 22, 1971 KJR

SPEED VARIATOR
ERIE, PA.

36C762868AA

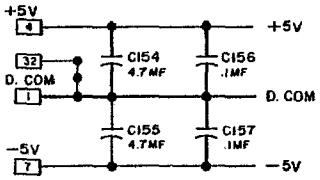
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

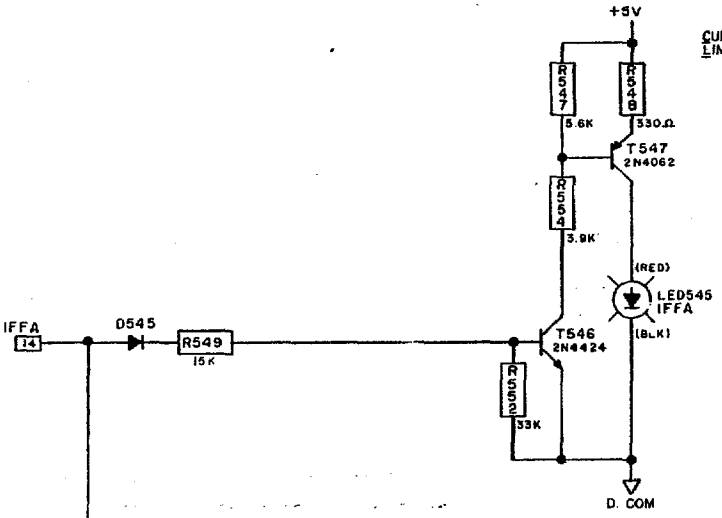
GENERAL ELECTRIC 36C762854AA

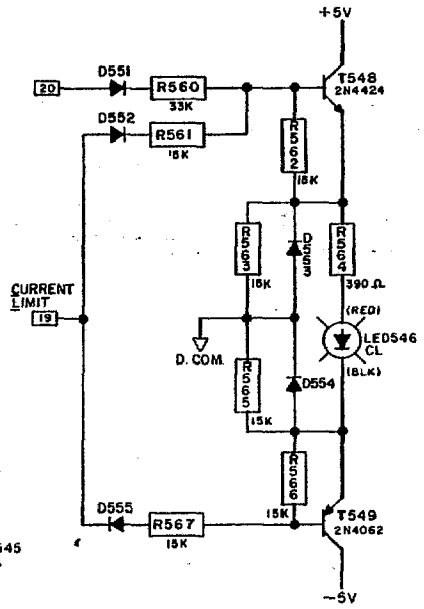
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING.—

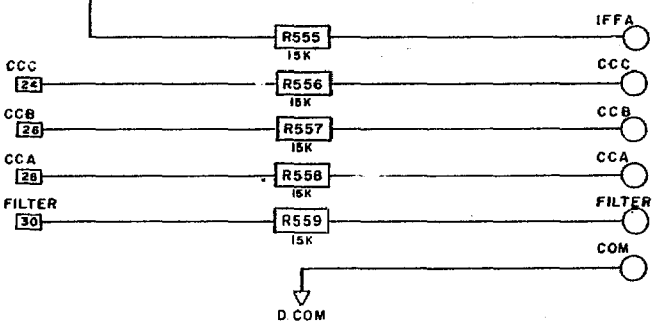
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

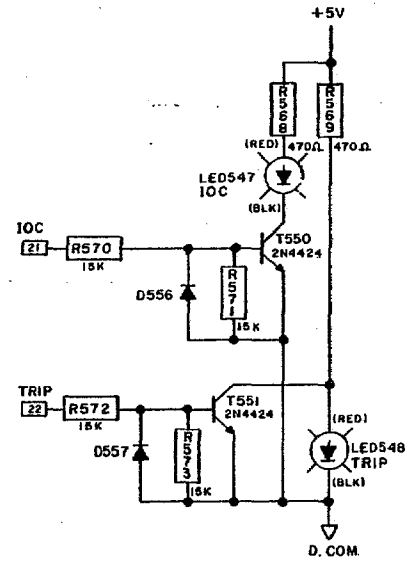
TITLE PRINTED CIRCUIT DIAGRAM
MONITOR
FIRST MADE FOR STANDARD LINE
193X184AAG01





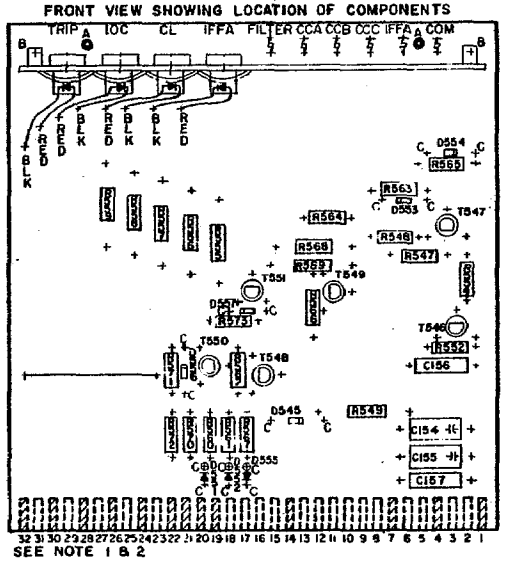






○ - INDICATES TEST CONNECTOR

FRONT VIEW SHOWING LOCATION OF COMPONENTS



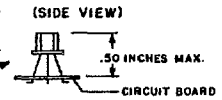
SEE NOTE 1 & 2

HOLE TABULATION

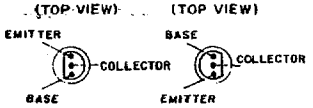
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.078	2
C	.032	16

NOTE 1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATE TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN HERE



TRANSISTOR LEAD SKETCHES



SKETCH FOR T548, T549 | SKETCH FOR T550 | SKETCH FOR T546, T551 | SKETCH FOR T547

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	9428526AA004	1	0

REVISIONS

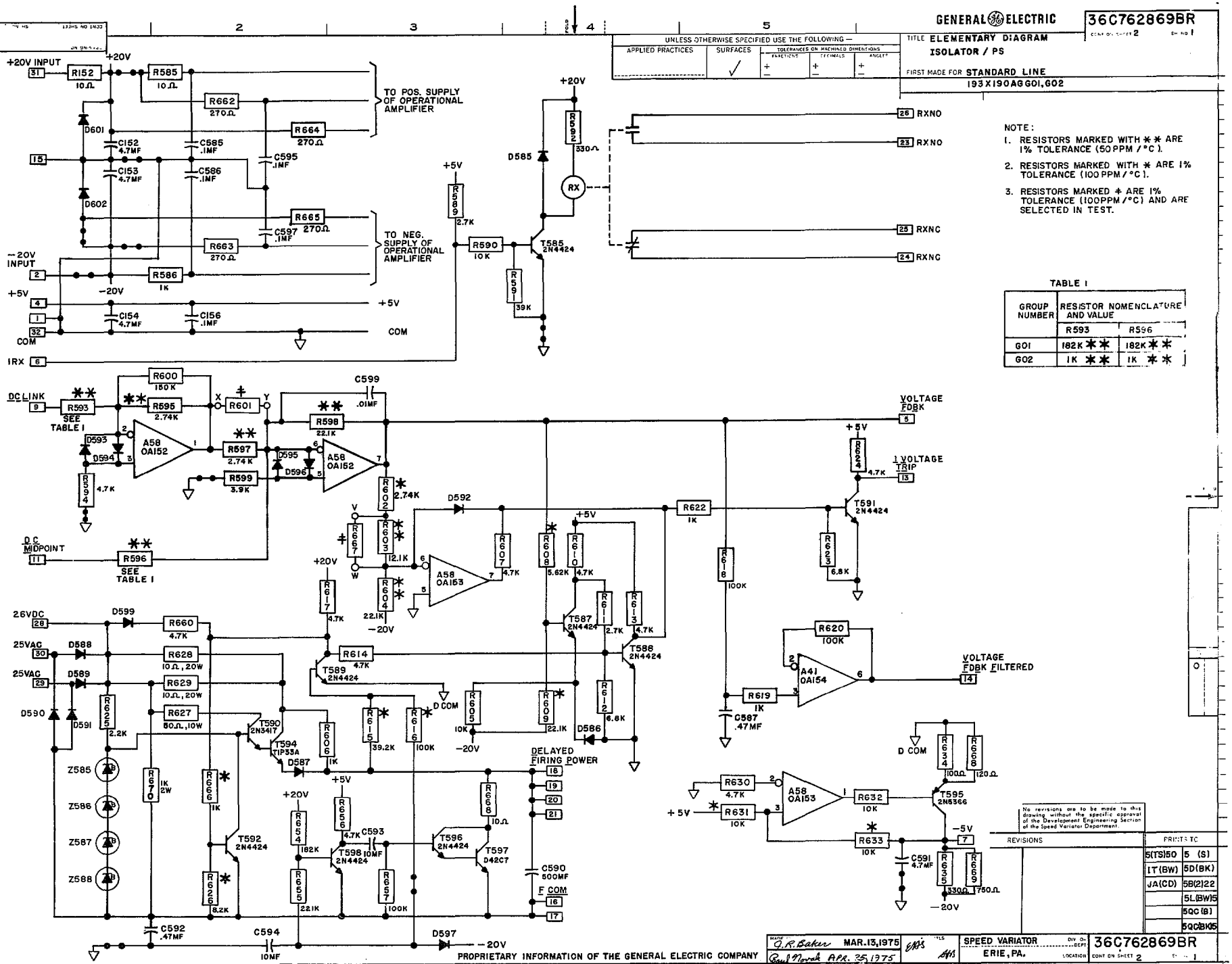
PRINTS

5QC(BK)	5 (S)
	5D(BK)
	5B(2)
	5(TS)5Q
	5(QC)B

MADE BY	APPROVALS	
G. Britton JUNE 3, 1970	GAS	SPEED VARIATOR
ISSUED K. Jensen FEB 15 1971	KPL	ERIE, PENNA.

36C762854AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C762869BR
TITLE ELEMENTARY DIAGRAM
ISOLATOR / PS
FIRST MADE FOR STANDARD LINE
193X190AGG01,G02
+20V INPUT
-20V INPUT
+5V
COM
IRX
TO POS. SUPPLY OF OPERATIONAL AMPLIFIER
TO NEG. SUPPLY OF OPERATIONAL AMPLIFIER
RXNO
RXNC
NOTE:
1. RESISTORS MARKED WITH ** ARE 1% TOLERANCE (50 PPM / °C).
2. RESISTORS MARKED WITH * ARE 1% TOLERANCE (100 PPM / °C).
3. RESISTORS MARKED + ARE 1% TOLERANCE (100PPM / °C) AND ARE SELECTED IN TEST.
TABLE I
GROUP NUMBER
RESISTOR NOMENCLATURE AND VALUE
R593
R596
G01
182K**
182K**
G02
1K**
1K**
DC LINK
SEE TABLE I
DC MIDPOINT
VOLTAGE FDBK
VOLTAGE TRIP
VOLTAGE FDBK FILTERED
26VDC
25VAC
25VAC
DELAYED FIRING POWER
D COM
F COM
-5V
-20V
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
REVISIONS
PRINTS TO
5(TS)50 5 (S)
1T(BW) 5D(BK)
JA(CD) 5B(2)22
5L(BW)15
5QC(B)
5QC(BK)05
SPEED VARIATOR
ERIE, PA.
36C762869BR
MAR.13,1975
APR. 25,1975
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
71GA515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
ISOLATOR / PS

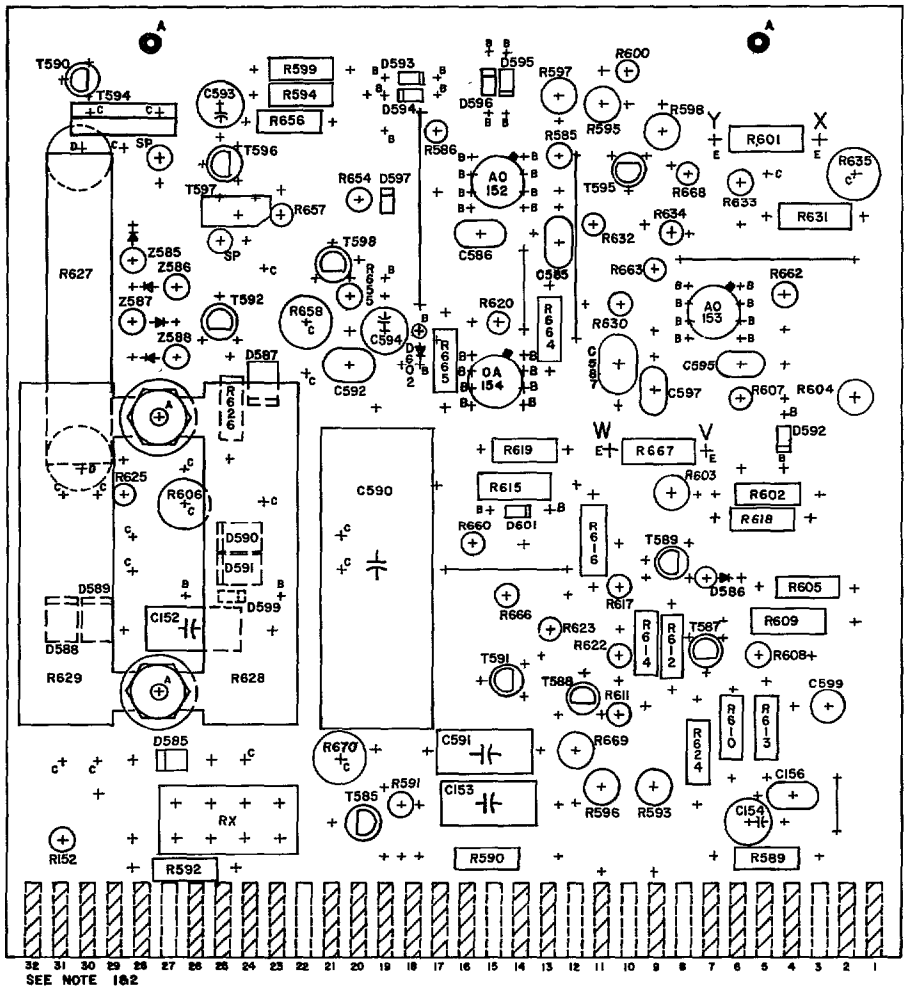
FIRST MADE FOR STANDARD LINE
193X190AGG01, G02

36C762869BR
CONT ON SHEET FL SH NO 2

HOLE TABULATION

ALL HOLES 040 DIA EXCEPT THE HOLES TABULATED BELOW

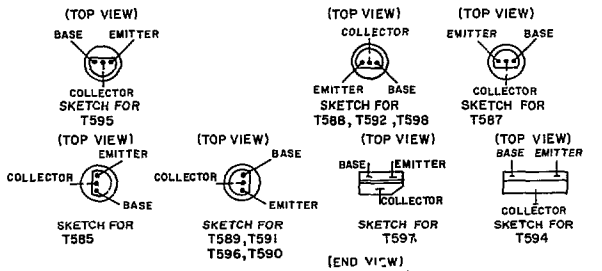
LOC	DIA	QUAN
A	.157	4
B	.032	42
C	.062	21
D	052	2
E	.078	4

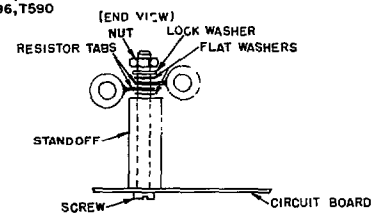


NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES





SKETCH FOR MOUNTING
R628, R629

2X	36A3529698D004		2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5QC(B) 5	(S)
5QC(BK) 5	5D(BK)
	5B(2)22
	5T3(5Q)

REV NO
Q

MADE BY	APPROVALS			
J R Baker MAR. 13, 1975	EPS	SPEED VARIATOR	DIV OR DEPT	36C762869BR
ISSUED Ken Novak APR 25, 1975	GHS	ERIE	LOCATION	CONT ON SHEET FL SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C762870AB

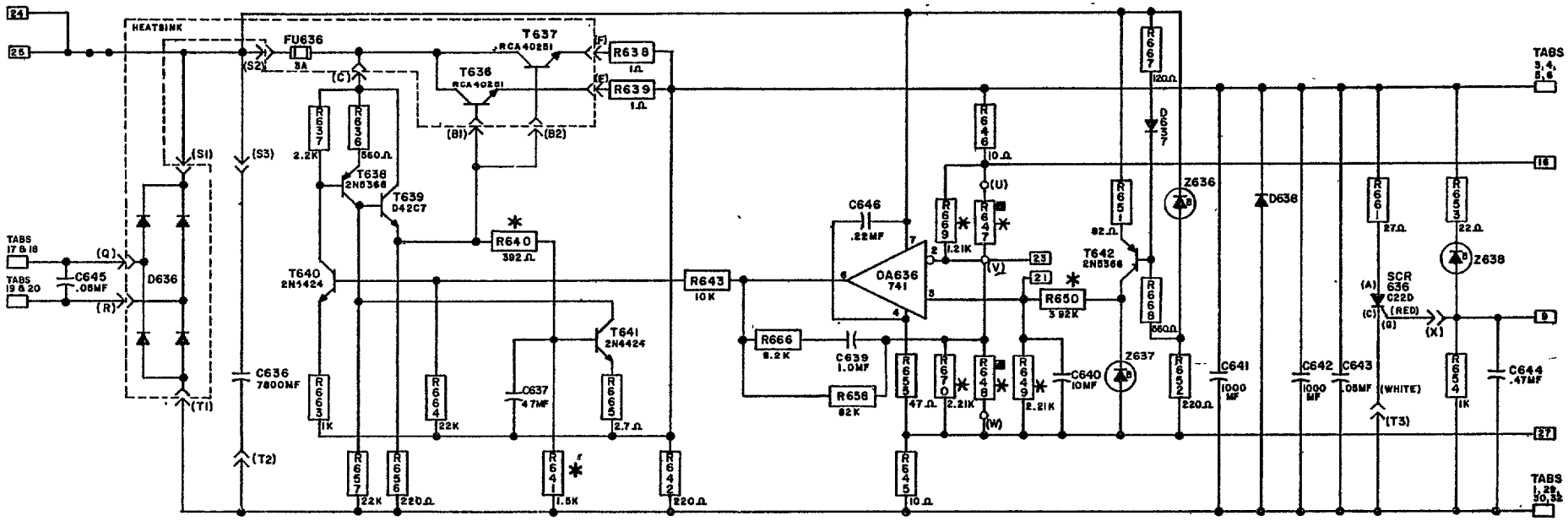
CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

TITLE ELEMENTARY DIAGRAM
+5 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X191AB G01



✱ INDICATE 1% RESISTOR

■ RESISTORS SELECTED IN TEST

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO	
5QC(BK) 5	5(S)
IT(BW)	5D(BK)
JA(CD)	5B(2)22
5AE(BW)	5L(BW)5
	5(TS)50
	5(QC) B

REV NO
0

DRAWN O. Mazza AUG. 25, 1973
TRACED B. Lasher FEB. 1, 1974
APPROVALS JT AB

SPEED VARIATOR ERIE, PA.

DIV OR DEPT 36C762870AB

LOCATION CONT ON SHEET 2 SH NO 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

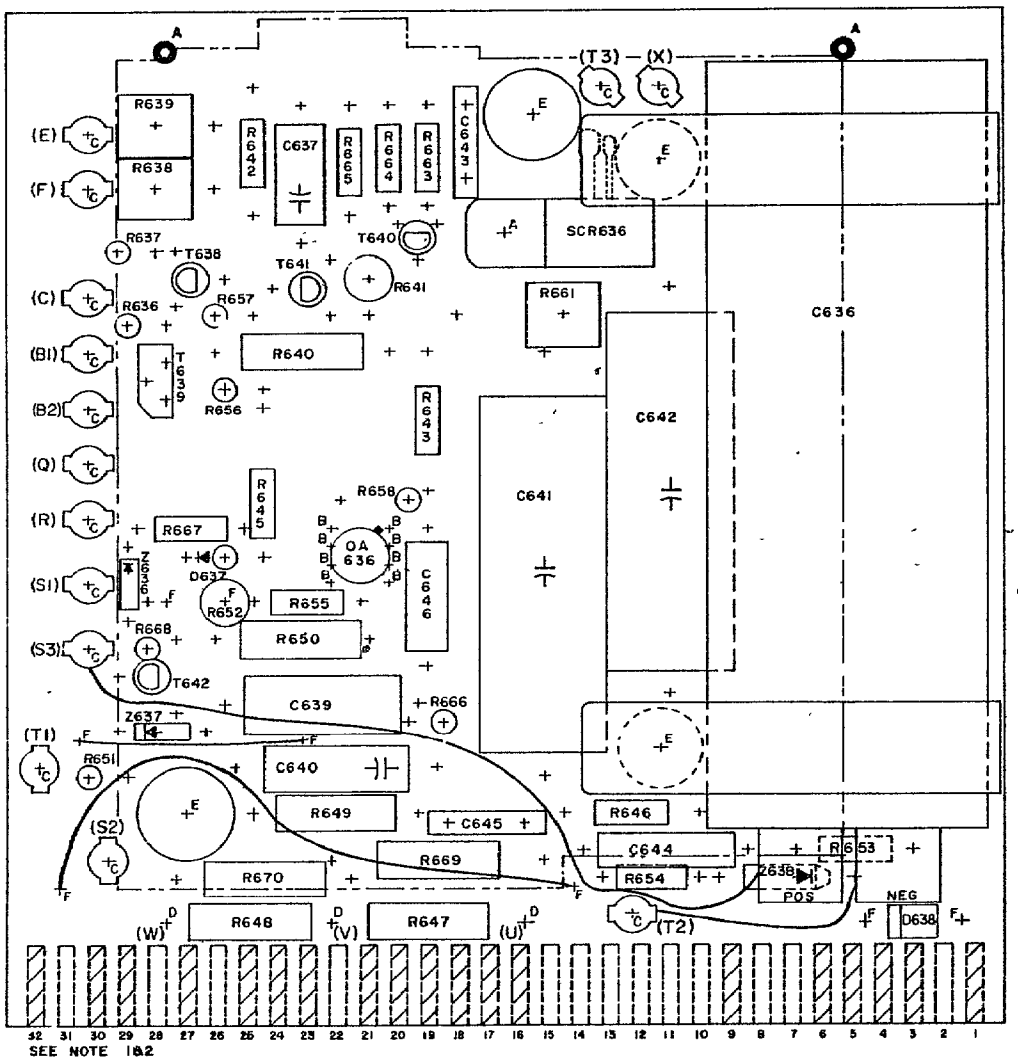
GENERAL ELECTRIC

36C762870AB

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	√	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
+ 5 VOLT POWER SUPPLY
FIRST MADE FOR STANDARD LINE
193X191AB G01



HOLE TABULATION

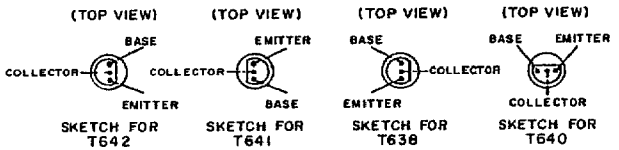
ALL HOLES 040 DIA. EXCEPT THE HOLES TABULATED BELOW

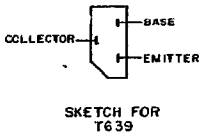
LOC.	DIA.	QUAN
A	.157	3
B	.032	8
C	101	14
D	.078	3
E	187	4
F	.062	8

NOTES

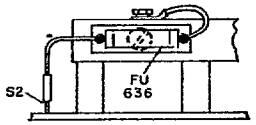
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.010}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES

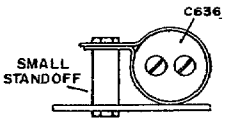




FUSE HOLDER CONNECTION SKETCH (SIDE VIEW OF HEATSINK FROM TAB END OF CARD)



CAPACITOR MOUNTING SKETCH



LEAD CONNECTIONS

CONNECT RING TERMINAL OF LEAD S3 TO POSITIVE TERMINAL OF C636
CONNECT RING TERMINAL OF LEAD T2 TO NEGATIVE TERMINAL OF C636
COUPLE RED LEAD FEMALE CONNECTOR TO MALE CONNECTOR X AND SCR636 WHITE LEAD FEMALE CONNECTOR TO MALE CONNECTOR T3
COUPLE REMAINING LEADS WITH FEMALE CONNECTORS TO MALE CONNECTORS BEARING NOMENCLATURE IDENTICAL TO THE LEAD MARKING

2X	36A352968AB004	3	0
SCALE	ETCHED CIRCUIT BOARD DWG.	FRONT SPACE	BACK UNIT REQUIREMENT

REVISIONS

PRINTS TO

5L(BW)5	5 (S)
1T(BW)	5D(BK)
JA(CD)	5B(2)22
5AE(BW)	5T8(60)
	5QC(BK)5
	5QC(B)

MADE BY A. Metta SEPT 4, 1973
APPROVED C.P. FEB 1 1974
SPEED VARIATOR
ERIE, PA
36C762870AB
CONT ON SHEET FL SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C762896AB

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

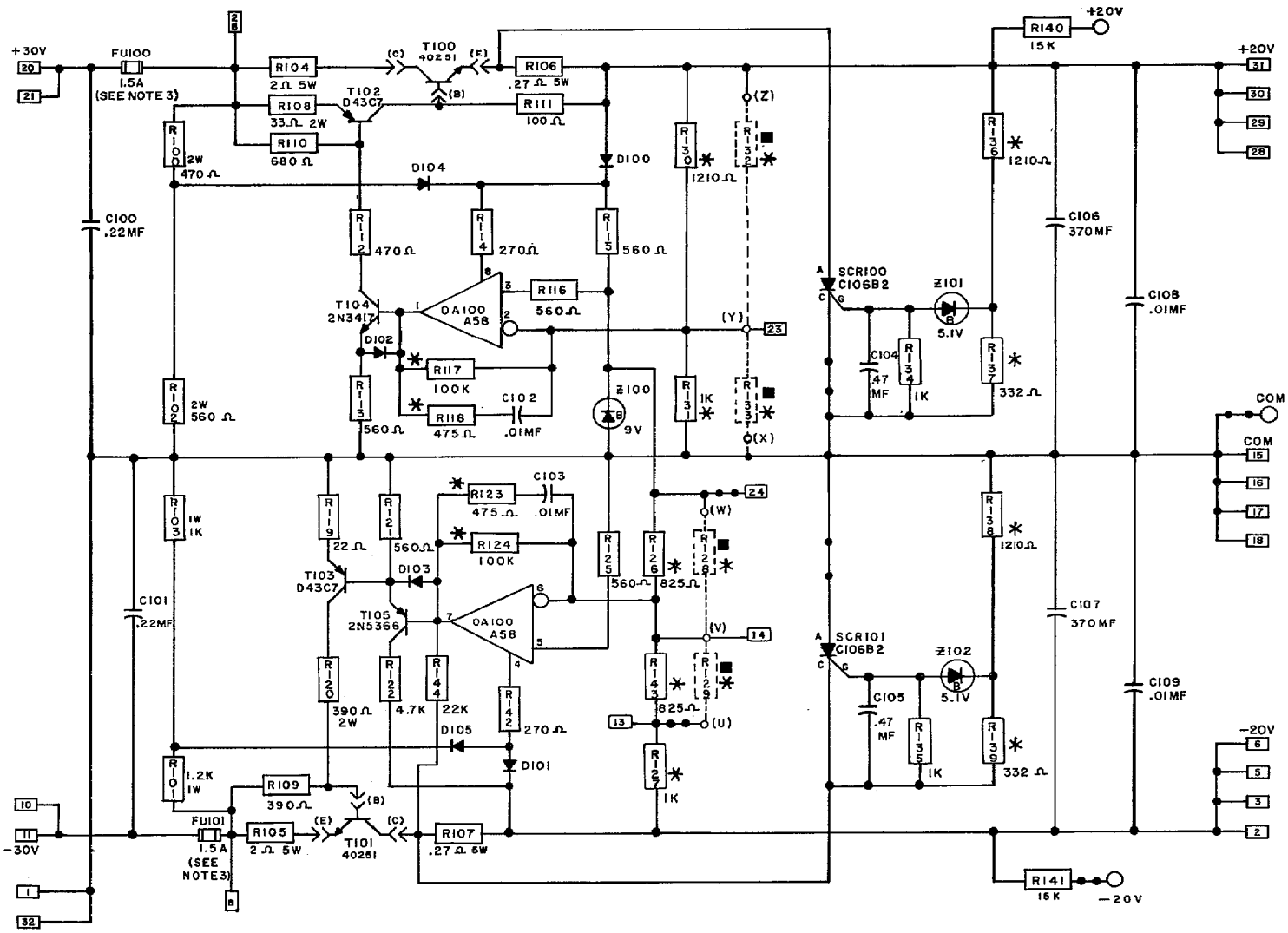
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A505	✓	±	±	±

TITLE ELEMENTARY DIAGRAM

20 VOLT POWER SUPPLY CARD

FIRST MADE FOR STANDARD LINE

193X257AAG01



NOTES 1. ✱ — 1% TEMPERATURE COMPENSATED RESISTOR.
■ — RESISTOR SELECTION IN TEST.

2. NUMBERS INSIDE SMALL RECTANGLES INDICATE TAB NUMBERS WHICH CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

3. FUSES ARE INSTRUMENT TYPE, LITTLE FUSE 361001, OR EQUIVALENT.

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

REV NO. O

PRINTS TO

JA(CD)	5(S)
IT(BW)	5Q(BK) 5
5L(BW)5	5D(BK)
5AE(BW)	5B(2)22
	5(T)50
	5(QC)BR

MADE BY C.R. Baker MARCH 15, 1974
CHKD BY G. Mourk SEPT. 4, 1974
APPROVALS dag

speed variator DIV OR DEPT
ERIE, PA. LOCATION

36C762896AB CONT ON SHEET 2 SH NO 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

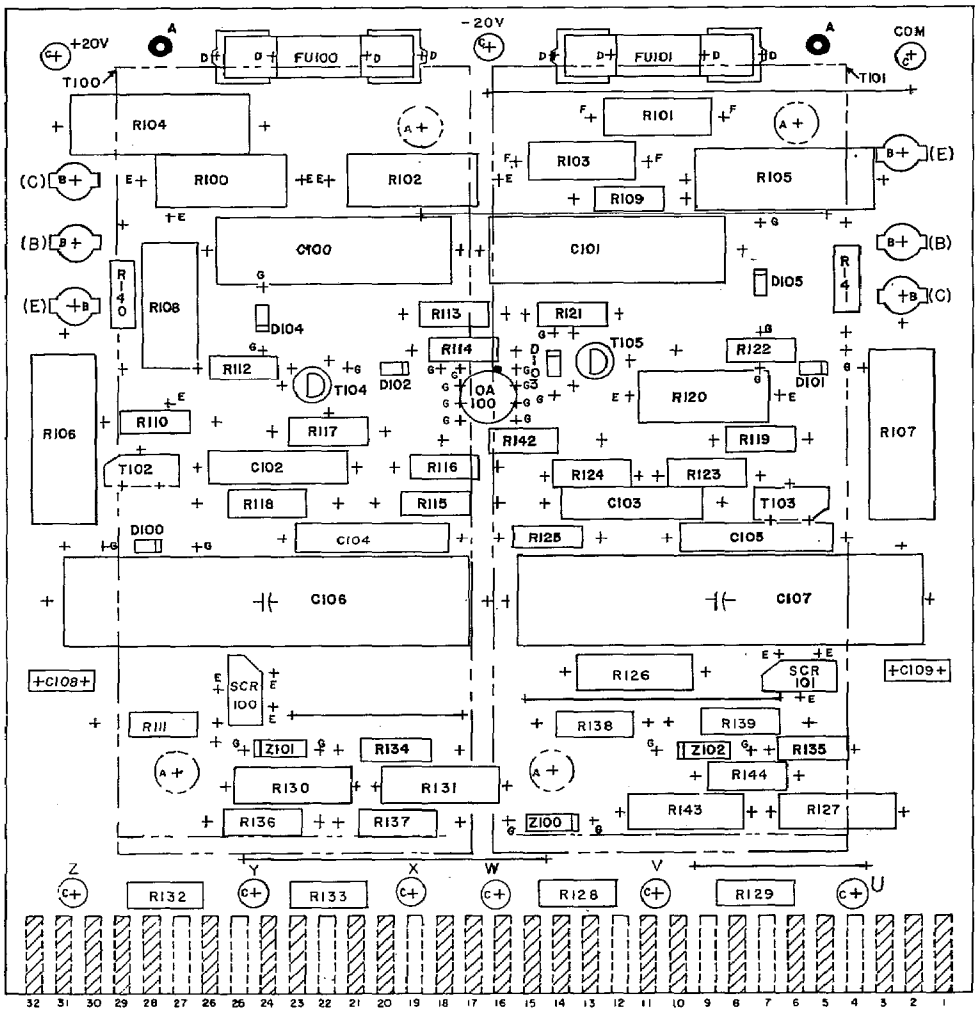
GENERAL ELECTRIC

36C762896AB

PRINTED CIRCUIT DIAGRAM
20 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X257AA801

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
APPLIED PRACTICES: 718A515 | SURFACES: ✓ | TOLERANCES IN MACHINE DIMENSIONS: FOR 1 PLACE ± | FOR 2 PLACE ± | FOR 3 PLACE ±



GROUP	KEY LOCATIONS			
01	6-7	12-13	26-27	

HOLE TABULATION

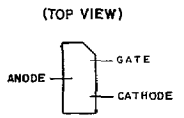
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	6
B	.101	6
C	.081	9
D	.078	8
E	.062	14
F	.052	4
G	.032	26

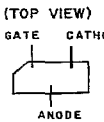
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

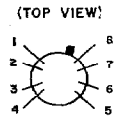
SCR LEAD SKETCHES



SKETCH FOR SCR100

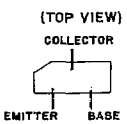


SKETCH FOR SCR101

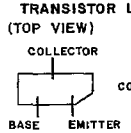


SKETCH FOR 0A100

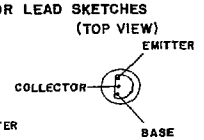
TRANSISTOR LEAD SKETCHES



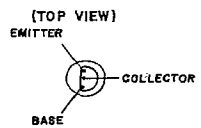
SKETCH FOR T102



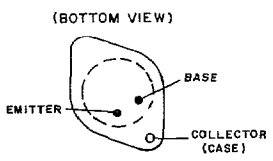
SKETCH FOR T103



SKETCH FOR T104



SKETCH FOR T105



SKETCH FOR T100, T101

2X	36A353740AB004		2		
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK	

REVISIONS

PHIL. S.I.
JA(CD) 5 (S)
IT(BW) 5D(BK)
5L(BW) 5B(2)22
5AE(BW) 5QQ(BK)5
5(TS)50
5QC(B)

MADE BY G. L. Baker MARCH 15, 1974
APPROVED Sept. 4, 1974

speed variator
ERIE, PA.

36C762896AB FL 2

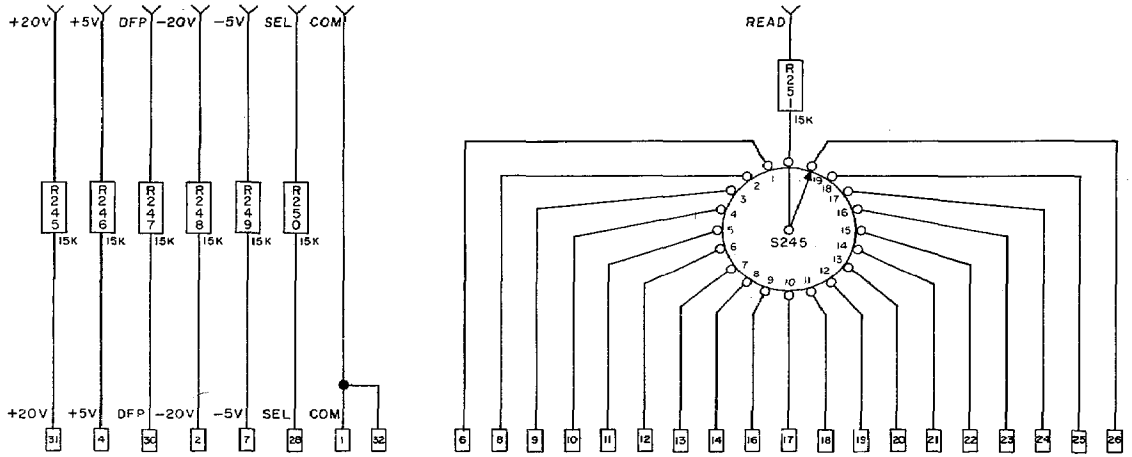
THE PROPERTY OF THE GENERAL ELECTRIC COMPANY

SCHEMATIC DIAGRAM

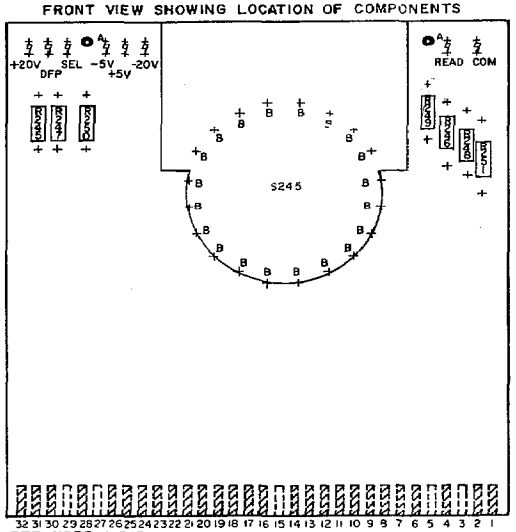
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM TEST CARD

FIRST MADE FOR STANDARD LINE 193X188AAG01



FRONT VIEW SHOWING LOCATION OF COMPONENTS



SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	20

NOTE 1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

2. CROSS HATCHED TABS INDICATE TABS USED.

3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

36C762858AA REV. 0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	9428549AA004	2	0

REVISIONS

PRINTS TO

5QC(BK) 5B(S)
5D(BK
5B(2
5(TS)5
5(QC)

MADE BY C.R. Baker NOV. 24, 1970
ISSUED JAN 29, 1971
APPROVALS Engs KXP
speed variator ERIE, PA.
DIV OR DEPT. LOCATION 36C762858AA CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

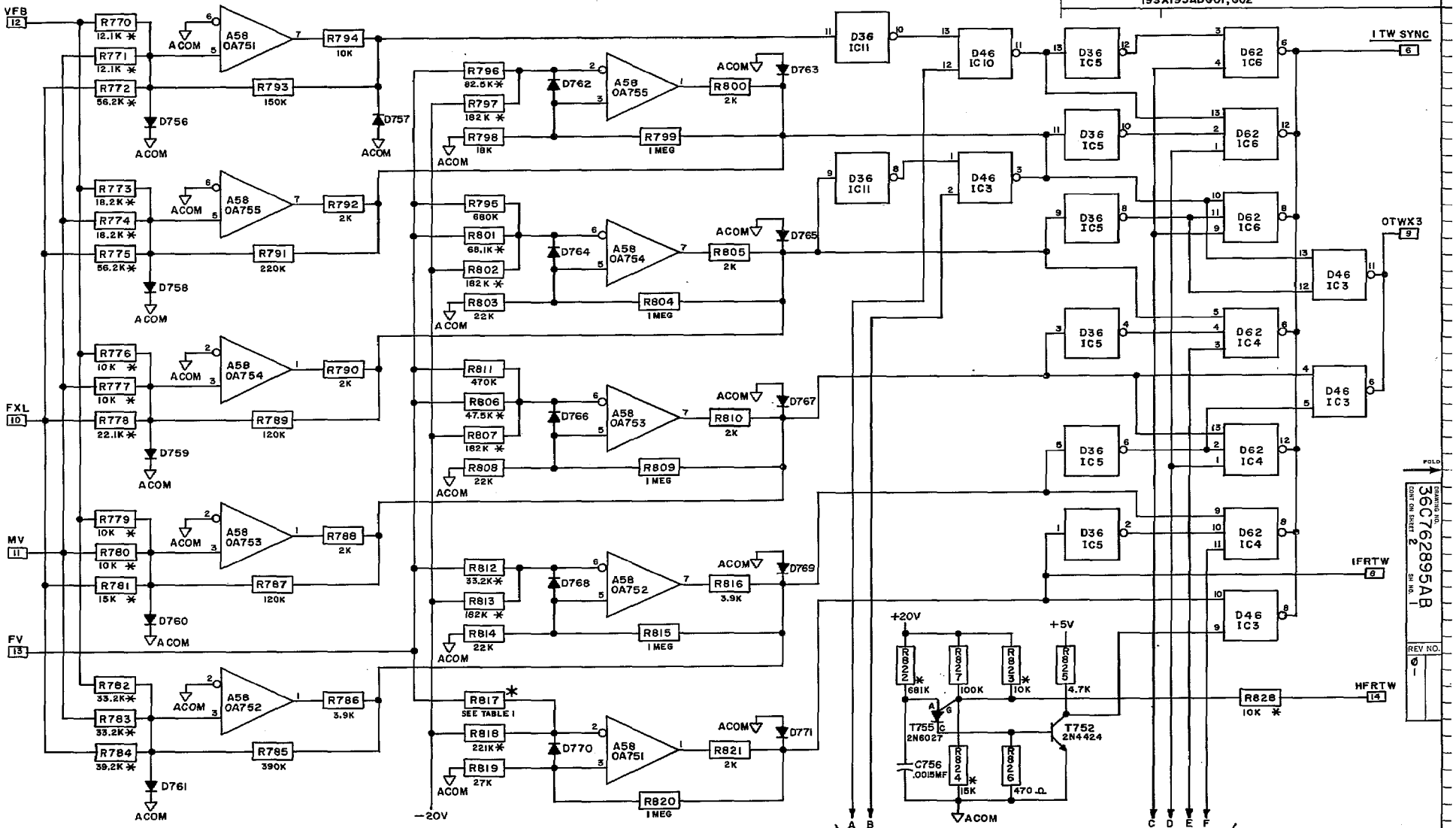


TABLE I

GROUP NO.	RESISTOR NO. AND VALUE
	R817
G01	10K
G02	5.62K

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

GENERAL ELECTRIC

36C762895AB CONT ON SHEET 2 SH NO. 1

TITLE ELEMENTARY DIAGRAM
DIVIDER CARD

FIRST MADE FOR STANDARD LINE
193X195ABG01, G02

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

DRAWING NO. 36C762895AB CONT ON SHEET 1 SH NO. 2

REV NO. 6

REVISIONS
6 M.B. Poppe 08 NOV. 28, 1978 BHB

PRINTS TO
5B(8)M 5J(BW)
5R(2)BW TS(50)
5L(2)BW
AW(BW)
5D(CD)
JA(CD)

MADE BY O. Platten SEPT 23, 1974
ISSUED W. Rajaraman SEPT 24, 1974
APPROVALS BHB M.B.
SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION
36C762895AB CONT ON SHEET 2 SH NO. 1

FP-6053-P (5-71)

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
DIVIDER CARD

FIRST MADE FOR STANDARD LINE
193X195AB G01,G02

+5V [4] — C154 4.7MF — C156 .1MF — +5V
D COM [1] [32] — D COM
+20V [31] — +20V — PIN 8 OF ALL OP. AMPS
C152 4.7MF
A COM [15] — C153 4.7MF — C753 .1MF — A COM
R753 56 Ω
R754 56 Ω — PIN 4 OF ALL OP. AMPS
−20V [2] — −20V

TO SH. 2 (A, B)
TO SH. 1 (C, D, E, F)

+5V — R755 1.5K — D36 IC9 — T751 2N4424
IFP [25] — R756 4.7K — R757 3.9K — D COM

D53 IC7 — D53 IC7 — D53 IC8 — D53 IC8
D36 IC9 — D46 IC10 — D36 IC9 — D46 IC10 — D36 IC9 — D46 IC10
D53 IC1 — D45 IC2 — D53 IC1
D36 IC11 — D36 IC11 — D36 IC9 — D36 IC11 — D36 IC9 — D36 IC11

+5V — R767 4.7K — R768 4.7K — R769 4.7K

BFX4 [24]
BFX2 [23]
BFXI [22]

[21] OFP/16
[20] OFP/8
[29] OFP/4
[28] OFP/2
[17] OFFX
[16] IFFX
[30] OFF6
[26] OFFY
[18] IFFY
[27] OFFZ
[19] IFFZ

DRAWING NO. 36C762895AB CONT ON SHEET 3 SH NO. 2
REV NO. Ø1

REVISIONS
1 | 71.0 Purple DS | NOV. 26, 1974 | BHB

PRINTS TO
5B(8)M | 5J(BW)
5R(2)BW | T5(5Ø)
5L(2)BW
AW(BW)
5D(CD)
JA(CD)

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

MADE BY D. Mattio SEPT 23,1974
ISSUED W.J. Kayanaugh SEPT 24,1974
APPROVALS BAS
SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION
36C762895AB CONT ON SHEET 3 SH NO. 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-608-P 10-71 PRINTED IN U.S.A.

GENERAL ELECTRIC

36C762895AB
CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

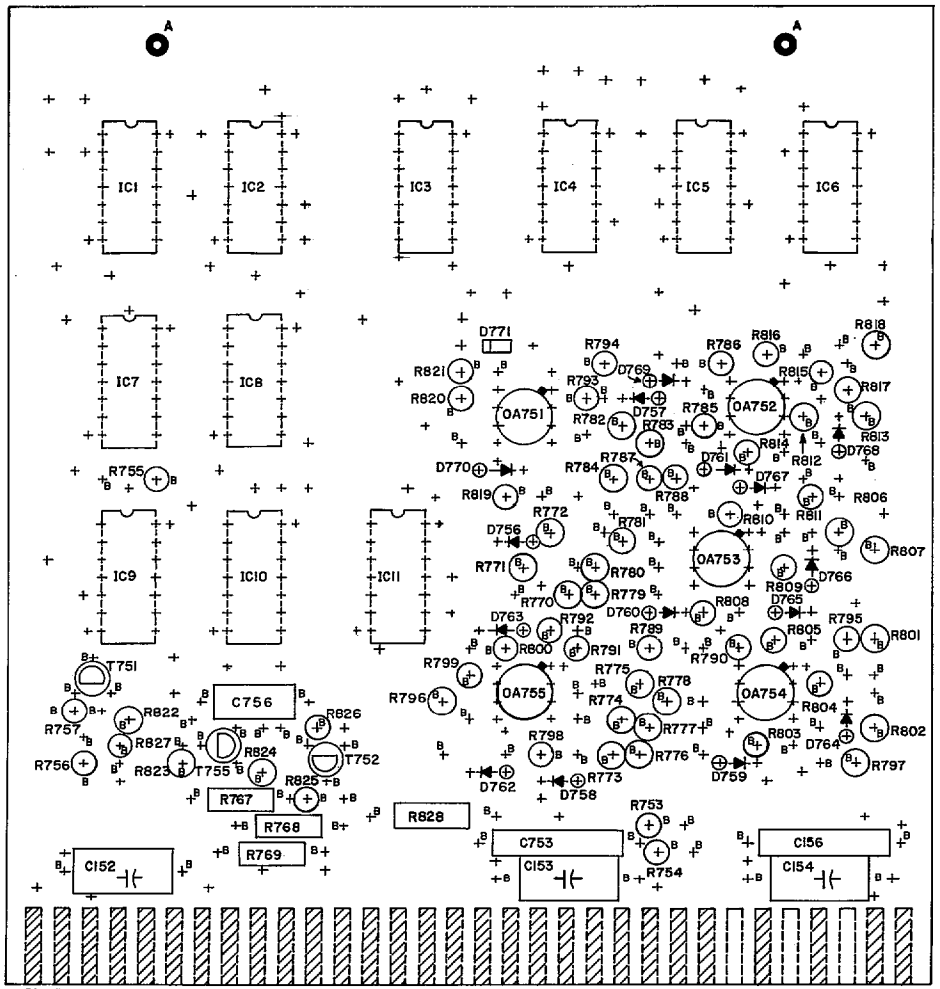
TITLE PRINTED CIRCUIT DIAGRAM
DIVIDER CARD

FIRST MADE FOR STANDARD LINE
193X195AB G01, G02

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

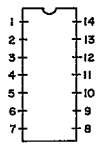
LOC.	DIA.	QUAN.
A	.157	2
B	.040	155



NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

(TOP VIEW)



SKETCH FOR IC PACKAGES

(TOP VIEW)



SKETCH FOR 0A751, 0A752, 0A753, 0A754, 0A755

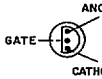
TRANSISTOR LEAD SKETCHES

(TOP VIEW)



SKETCH FOR T752

(TOP VIEW)



SKETCH FOR T755

(TOP VIEW)



SKETCH FOR T751

2X	36A353738AA005	36A353739AA005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV NO.			
Ø	1	718 Purple #5 NOV 29, 1978	

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	T5(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY	DATE	APPROVALS		DIV. OR DEPT	DRAWING NO.
D Morris	SEPT 23, 1974	EMS		SPEED VARIATOR	36C762895AB
D Thayanandan	SEPT 24 1974	MD		ERIE LOCATION	CONT ON SHEET FL SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

SCHEMATIC DIAGRAM

GENERAL ELECTRIC

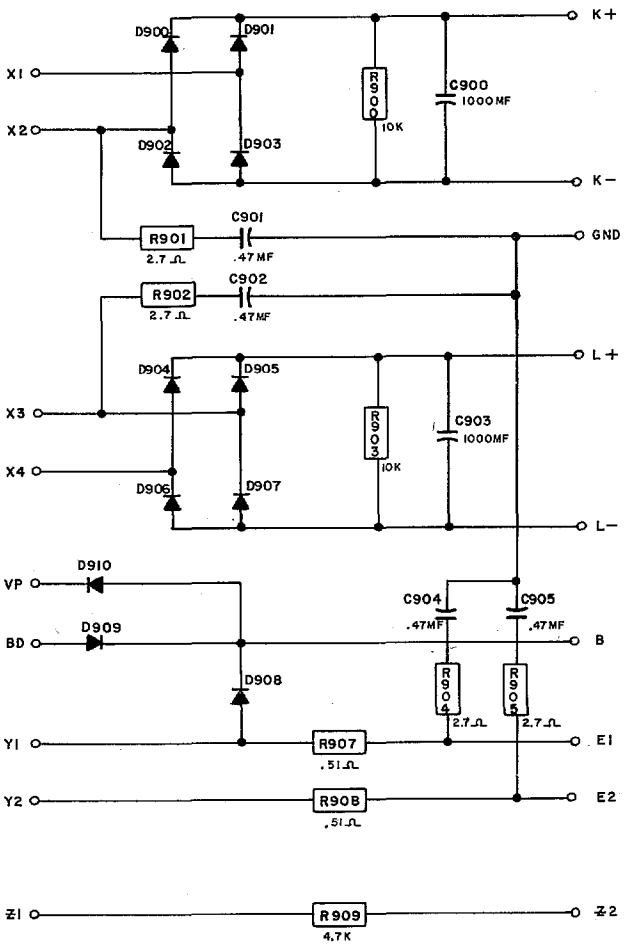
36C764129AA

CONT ON SHEET | SH NO.

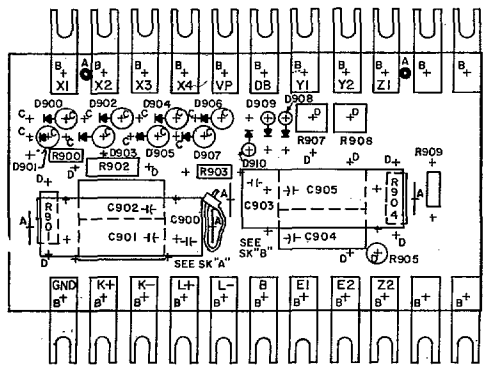
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
POWER INTERFACE
FIRST MADE FOR STANDARD LINE
193X370AA G01



FRONT VIEW SHOWING LOCATION OF COMPONENTS

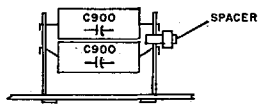


HOLE TABULATION

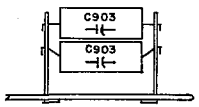
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	6
B	.101	22
C	.062	16
D	.052	12

CARD SIZE, $2.85^{\pm.020}$ X $5.130^{+.002}_{-.008}$



SKETCH "A"



SKETCH "B"

REV NO.
0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	36B605049AA004	———	———

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5QC(BK)5	5 (S)
IT(BW)	5D(BK)
	5(TS)50
	5B(2)22
	JA(CD)
	5(QC)B

MADE BY R. Babo JUNE 20, 1973
ISSUED R.D. Riley FEB. 1 1974

APPROVALS

SPEED VARIATOR ERIE, PA.

DIV OR DEPT | LOCATION

36C764129AA

CONT ON SHEET | SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764140AA

CONT ON SHEET

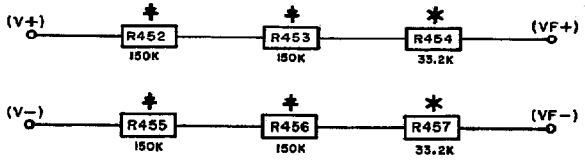
SH NO

SCHEMATIC DIAGRAM

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
RESISTANCE ISOLATOR CARD

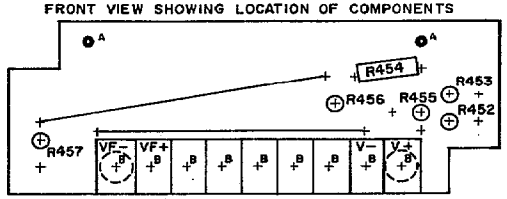
FIRST MADE FOR STANDARD LINE
193X263AAG03



NOTES

1. ✱ 1% RESISTOR — 50 PPM
2. ✚ .2% RESISTOR — 50 PPM

FRONT VIEW SHOWING LOCATION OF COMPONENTS



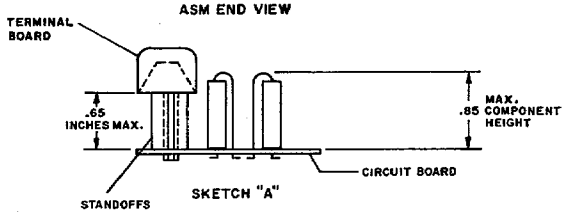
HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.078	9

NOTE—1. CARD SIZE 1.90 $^{+.000}_{-.015}$ X 5.130 $^{+.002}_{-.006}$

ASM END VIEW



SKETCH "A"

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMEN	
		FRONT	BACK
FULL	36B605027AA004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varidor Department.

PRINT

5QC(BK)5	JA(C
5B(2)22	5D(B
5(TS)50	IT(B
5(S)	5L(B
	5AE(B
	5(QC

MADE BY J.R. Baker DEC. 17, 1973
ISSUED Paul Novak MARCH 26, 1974

APPROVALS

speed varidor
ERIE, PA.

DIV OR DEPT

36C764140AA

LOCATION CONT ON SHEET SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY